Cisco 640-554

Implementing Cisco IOS Network Security (IINS v2.0)

Version: 5.0

QUESTION NO: 1

Which two features are supported by Cisco IronPort Security Gateway? (Choose two.)

A. spam protection
B. outbreak intelligence
C. HTTP and HTTPS scanning
D. email encryption
E. DDoS protection

Answer: A,D
Explanation:

QUESTION NO: 2

Which option is a feature of Cisco ScanSafe technology?

A. spam protection
B. consistent cloud-based policy
C. DDoS protection
D. RSA Email DLP

Answer: B
Explanation:

QUESTION NO: 3

Which two characteristics represent a blended threat? (Choose two.)

A. man-in-the-middle attack
B. trojan horse attack
C. pharming attack
D. denial of service attack
E. day zero attack

Answer: B,E
Explanation:

QUESTION NO: 4

Under which higher-level policy is a VPN security policy categorized?

A. application policy
B. DLP policy
C. remote access policy
D. compliance policy
E. corporate WAN policy

Answer: C
Explanation:

QUESTION NO: 5

Refer to the exhibit.

```
router#sh run | include username

username test secret 5 $1$knm.$GOGQBIL8TK77P0LWxvX4O0
```

What does the option secret 5 in the username global configuration mode command indicate about the user password?

A. It is hashed using SHA.
B. It is encrypted using DH group 5.
C. It is hashed using MD5.
D. It is encrypted using the service password-encryption command.
E. It is hashed using a proprietary Cisco hashing algorithm.
F. It is encrypted using a proprietary Cisco encryption algorithm.

Answer: C
Explanation:

QUESTION NO: 6

What does level 5 in this enable secret global configuration mode command indicate?

router#enable secret level 5 password

A. The enable secret password is hashed using MD5.
B. The enable secret password is hashed using SHA.
C. The enable secret password is encrypted using Cisco proprietary level 5 encryption.
D. Set the enable secret command to privilege level 5.
E. The enable secret password is for accessing exec privilege level 5.

Answer: E
Explanation:

QUESTION NO: 7

Which Cisco management tool provides the ability to centrally provision all aspects of device configuration across the Cisco family of security products?

A. Cisco Configuration Professional
B. Security Device Manager
C. Cisco Security Manager
D. Cisco Secure Management Server

Answer: C
Explanation:

QUESTION NO: 8

Which option is the correct representation of the IPv6 address 2001:0000:150C:0000:0000:41B1:45A3:041D?

A. 2001::150c::41b1:45a3:041d
B. 2001:0:150c:0::41b1:45a3:04d1
C. 2001:150c::41b1:45a3::41d
D. 2001:0:150c::41b1:45a3:41d

Answer: D
Explanation:

QUESTION NO: 9

Which three options are common examples of AAA implementation on Cisco routers? (Choose three.)

A. authenticating remote users who are accessing the corporate LAN through IPsec VPN connections
B. authenticating administrator access to the router console port, auxiliary port, and vty ports
C. implementing PKI to authenticate and authorize IPsec VPN peers using digital certificates
D. tracking Cisco NetFlow accounting statistics
E. securing the router by locking down all unused services
F. performing router commands authorization using TACACS+

Answer: A,B,F
Explanation:

QUESTION NO: 10

When AAA login authentication is configured on Cisco routers, which two authentication methods should be used as the final method to ensure that the administrator can still log in to the router in case the external AAA server fails? (Choose two.)

A. group RADIUS
B. group TACACS+
C. local
D. krb5
E. enable
F. if-authenticated

Answer: C,E
Explanation:

QUESTION NO: 11

Which two characteristics of the TACACS+ protocol are true? (Choose two.)

A. uses UDP ports 1645 or 1812
B. separates AAA functions
C. encrypts the body of every packet
D. offers extensive accounting capabilities
E. is an open RFC standard protocol

Answer: B,C
Explanation:

QUESTION NO: 12

Refer to the exhibit.

```
Oct 13 19:46:06.170: AAA/MEMORY: create_user (0x4C5E1F60) user='tecteam'
ruser='NULL' ds0=0 port='tty515' rem_addr='10.0.2.13' authen_type=ASCII
service=ENABLE priv=15 initial_task_id='0', vrf= (id=0)
Oct 13 19:46:06.170: AAA/AUTHEN/START (2600878790): port='tty515' list=''
action=LOGIN service=ENABLE
Oct 13 19:46:06.170: AAA/AUTHEN/START (2600878790): console enable - default to
enable password (if any)
Oct 13 19:46:06.170: AAA/AUTHEN/START (2600878790): Method=ENABLE
Oct 13 19:46:06.170: AAA/AUTHEN (2600878790): status = GETPASS
Oct 13 19:46:07.266: AAA/AUTHEN/CONT (2600878790): continue_login
(user='(undef)')
Oct 13 19:46:07.266: AAA/AUTHEN (2600878790): status = GETPASS
Oct 13 19:46:07.266: AAA/AUTHEN/CONT (2600878790): Method=ENABLE
Oct 13 19:46:07.266: AAA/AUTHEN(2600878790): password incorrect
Oct 13 19:46:07.266: AAA/AUTHEN (2600878790): status = FAIL
Oct 13 19:46:07.266: AAA/MEMORY: free_user (0x4C5E1F60) user='NULL'
ruser='NULL' port='tty515' rem_addr='10.0.2.13' authen_type=ASCII service=ENABLE
priv=15 vrf= (id=0)
```

Which statement about this output is true?

A. The user logged into the router with the incorrect username and password.
B. The login failed because there was no default enable password.
C. The login failed because the password entered was incorrect.
D. The user logged in and was given privilege level 15.

Answer: C
Explanation:

QUESTION NO: 13

Refer to the exhibit.

```
access-list 100 permit tcp 172.26.26.16 0.0.0.7 host 192.168.1.2 eq 443
access-list 100 permit tcp 172.26.26.16 0.0.0.7 host 192.168.1.2 eq 80
access-list 100 deny   tcp any host 192.168.1.2 eq telnet
access-list 100 deny   tcp any host 192.168.1.2 eq www
access-list 100 permit ip any any
```

Which traffic is permitted by this ACL?

A. TCP traffic sourced from any host in the 172.26.26.8/29 subnet on any port to host 192.168.1.2 port 80 or 443

B. TCP traffic sourced from host 172.26.26.21 on port 80 or 443 to host 192.168.1.2 on any port

C. any TCP traffic sourced from host 172.26.26.30 destined to host 192.168.1.1

D. any TCP traffic sourced from host 172.26.26.20 to host 192.168.1.2

Answer: C

Explanation:

QUESTION NO: 14

Refer to the exhibit.

```
access-list 2 permit 10.10.0.10
access-list 2 deny 10.10.0.0 0.0.255.255
access-list 2 permit 10.0.0.0 0.255.255.255
interface FastEthernet0/0
   ip access-group 2 in
```

Which statement about this partial CLI configuration of an access control list is true?

A. The access list accepts all traffic on the 10.0.0.0 subnets.

B. All traffic from the 10.10.0.0 subnets is denied.

C. Only traffic from 10.10.0.10 is allowed.

D. This configuration is invalid. It should be configured as an extended ACL to permit the associated wildcard mask.

E. From the 10.10.0.0 subnet, only traffic sourced from 10.10.0.10 is allowed; traffic sourced from the other 10.0.0.0 subnets also is allowed.

F. The access list permits traffic destined to the 10.10.0.10 host on FastEthernet0/0 from any source.

Answer: E

Explanation:

QUESTION NO: 15

Which type of Cisco ASA access list entry can be configured to match multiple entries in a single statement?

A. nested object-class
B. class-map
C. extended wildcard matching
D. object groups

Answer: D
Explanation:

QUESTION NO: 16

Which statement about an access control list that is applied to a router interface is true?

A. It only filters traffic that passes through the router.
B. It filters pass-through and router-generated traffic.
C. An empty ACL blocks all traffic.
D. It filters traffic in the inbound and outbound directions.

Answer: A
Explanation:

QUESTION NO: 17

You have been tasked by your manager to implement syslog in your network. Which option is an important factor to consider in your implementation?

A. Use SSH to access your syslog information.
B. Enable the highest level of syslog function available to ensure that all possible event messages are logged.
C. Log all messages to the system buffer so that they can be displayed when accessing the router.
D. Synchronize clocks on the network with a protocol such as Network Time Protocol.

Answer: D

Explanation:

QUESTION NO: 18

Which protocol secures router management session traffic?

A. SSTP
B. POP
C. Telnet
D. SSH

Answer: D

Explanation:

QUESTION NO: 19

Which two considerations about secure network management are important? (Choose two.)

A. log tampering
B. encryption algorithm strength
C. accurate time stamping
D. off-site storage
E. Use RADIUS for router commands authorization.
F. Do not use a loopback interface for device management access.

Answer: A,C

Explanation:

QUESTION NO: 20

Which command enables Cisco IOS image resilience?

A. secure boot-<IOS image filename>
B. secure boot-running-config
C. secure boot-start
D. secure boot-image

Answer: D

Explanation:

QUESTION NO: 21

Which router management feature provides for the ability to configure multiple administrative views?

A. role-based CLI
B. virtual routing and forwarding
C. secure config privilege {level}
D. parser view view name

Answer: A

Explanation:

QUESTION NO: 22

You suspect that an attacker in your network has configured a rogue Layer 2 device to intercept traffic from multiple VLANs, which allows the attacker to capture potentially sensitive data.

Which two methods will help to mitigate this type of activity? (Choose two.)

A. Turn off all trunk ports and manually configure each VLAN as required on each port.
B. Place unused active ports in an unused VLAN.
C. Secure the native VLAN, VLAN 1, with encryption.
D. Set the native VLAN on the trunk ports to an unused VLAN.
E. Disable DTP on ports that require trunking.

Answer: D,E

Explanation:

QUESTION NO: 23

Which statement describes a best practice when configuring trunking on a switch port?

A. Disable double tagging by enabling DTP on the trunk port.
B. Enable encryption on the trunk port.

C. Enable authentication and encryption on the trunk port.
D. Limit the allowed VLAN(s) on the trunk to the native VLAN only.
E. Configure an unused VLAN as the native VLAN.

Answer: E
Explanation:

QUESTION NO: 24

Which type of Layer 2 attack causes a switch to flood all incoming traffic to all ports?

A. MAC spoofing attack
B. CAM overflow attack
C. VLAN hopping attack
D. STP attack

Answer: B
Explanation:

QUESTION NO: 25

What is the best way to prevent a VLAN hopping attack?

A. Encapsulate trunk ports with IEEE 802.1Q.
B. Physically secure data closets.
C. Disable DTP negotiations.
D. Enable BDPU guard.

Answer: C
Explanation:

QUESTION NO: 26

Which statement about PVLAN Edge is true?

A. PVLAN Edge can be configured to restrict the number of MAC addresses that appear on a single port.
B. The switch does not forward any traffic from one protected port to any other protected port.

C. By default, when a port policy error occurs, the switchport shuts down.
D. The switch only forwards traffic to ports within the same VLAN Edge.

Answer: B
Explanation:

QUESTION NO: 27

If you are implementing VLAN trunking, which additional configuration parameter should be added to the trunking configuration?

A. no switchport mode access
B. no switchport trunk native VLAN 1
C. switchport mode DTP
D. switchport nonnegotiate

Answer: D
Explanation:

QUESTION NO: 28

When Cisco IOS zone-based policy firewall is configured, which three actions can be applied to a traffic class? (Choose three.)

A. pass
B. police
C. inspect
D. drop
E. queue
F. shape

Answer: A,C,D
Explanation:

QUESTION NO: 29

With Cisco IOS zone-based policy firewall, by default, which three types of traffic are permitted by the router when some of the router interfaces are assigned to a zone? (Choose three.)

A. traffic flowing between a zone member interface and any interface that is not a zone member
B. traffic flowing to and from the router interfaces (the self zone)
C. traffic flowing among the interfaces that are members of the same zone
D. traffic flowing among the interfaces that are not assigned to any zone
E. traffic flowing between a zone member interface and another interface that belongs in a different zone
F. traffic flowing to the zone member interface that is returned traffic

Answer: B,C,D
Explanation:

QUESTION NO: 30

Which option is a key difference between Cisco IOS interface ACL configurations and Cisco ASA appliance interface ACL configurations?

A. The Cisco IOS interface ACL has an implicit permit-all rule at the end of each interface ACL.
B. Cisco IOS supports interface ACL and also global ACL. Global ACL is applied to all interfaces.
C. The Cisco ASA appliance interface ACL configurations use netmasks instead of wildcard masks.
D. The Cisco ASA appliance interface ACL also applies to traffic directed to the IP addresses of the Cisco ASA appliance interfaces.
E. The Cisco ASA appliance does not support standard ACL. The Cisco ASA appliance only support extended ACL.

Answer: C
Explanation:

QUESTION NO: 31

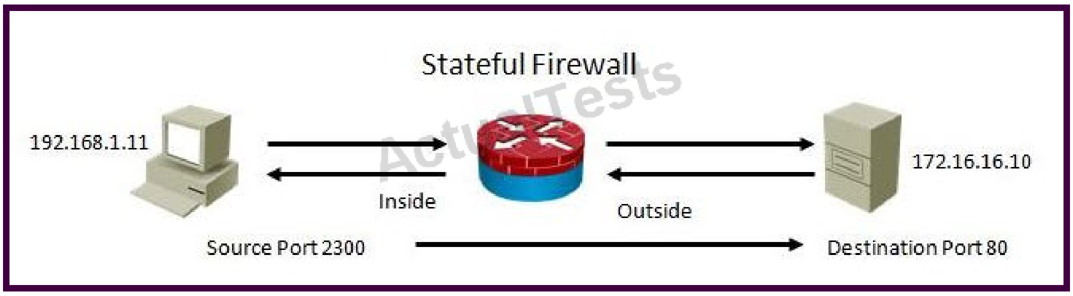
Which two options are advantages of an application layer firewall? (Choose two.)

A. provides high-performance filtering
B. makes DoS attacks difficult
C. supports a large number of applications
D. authenticates devices
E. authenticates individuals

Answer: B,E
Explanation:

QUESTION NO: 32

Refer to the exhibit.



Using a stateful packet firewall and given an inside ACL entry of permit ip 192.16.1.0 0.0.0.255 any, what would be the resulting dynamically configured ACL for the return traffic on the outside ACL?

A. permit tcp host 172.16.16.10 eq 80 host 192.168.1.11 eq 2300
B. permit ip 172.16.16.10 eq 80 192.168.1.0 0.0.0.255 eq 2300
C. permit tcp any eq 80 host 192.168.1.11 eq 2300
D. permit ip host 172.16.16.10 eq 80 host 192.168.1.0 0.0.0.255 eq 2300

Answer: A
Explanation:

QUESTION NO: 33

Which option is the resulting action in a zone-based policy firewall configuration with these conditions?

Source: Zone 1

Destination: Zone 2

Zone pair exists?: Yes

Policy exists?: No

A. no impact to zoning or policy
B. no policy lookup (pass)

C. drop
D. apply default policy

Answer: C
Explanation:

QUESTION NO: 34

A Cisco ASA appliance has three interfaces configured. The first interface is the inside interface with a security level of 100. The second interface is the DMZ interface with a security level of 50. The third interface is the outside interface with a security level of 0.

By default, without any access list configured, which five types of traffic are permitted? (Choose five.)

A. outbound traffic initiated from the inside to the DMZ
B. outbound traffic initiated from the DMZ to the outside
C. outbound traffic initiated from the inside to the outside
D. inbound traffic initiated from the outside to the DMZ
E. inbound traffic initiated from the outside to the inside
F. inbound traffic initiated from the DMZ to the inside
G. HTTP return traffic originating from the inside network and returning via the outside interface
H. HTTP return traffic originating from the inside network and returning via the DMZ interface
I. HTTP return traffic originating from the DMZ network and returning via the inside interface
J. HTTP return traffic originating from the outside network and returning via the inside interface

Answer: A,B,C,G,H
Explanation:

QUESTION NO: 35

Which two protocols enable Cisco Configuration Professional to pull IPS alerts from a Cisco ISR router? (Choose two.)

A. syslog
B. SDEE
C. FTP
D. TFTP
E. SSH
F. HTTPS

Answer: B,F

Explanation:

QUESTION NO: 36

Which two functions are required for IPsec operation? (Choose two.)

A. using SHA for encryption
B. using PKI for pre-shared key authentication
C. using IKE to negotiate the SA
D. using AH protocols for encryption and authentication
E. using Diffie-Hellman to establish a shared-secret key

Answer: C,E

Explanation:

QUESTION NO: 37

On Cisco ISR routers, for what purpose is the realm-cisco.pub public encryption key used?

A. used for SSH server/client authentication and encryption
B. used to verify the digital signature of the IPS signature file
C. used to generate a persistent self-signed identity certificate for the ISR so administrators can authenticate the ISR when accessing it using Cisco Configuration Professional
D. used to enable asymmetric encryption on IPsec and SSL VPNs
E. used during the DH exchanges on IPsec VPNs

Answer: B

Explanation:

QUESTION NO: 38

Which four tasks are required when you configure Cisco IOS IPS using the Cisco Configuration Professional IPS wizard? (Choose four.)

A. Select the interface(s) to apply the IPS rule.
B. Select the traffic flow direction that should be applied by the IPS rule.
C. Add or remove IPS alerts actions based on the risk rating.

D. Specify the signature file and the Cisco public key.
E. Select the IPS bypass mode (fail-open or fail-close).
F. Specify the configuration location and select the category of signatures to be applied to the selected interface(s).

Answer: A,B,D,F
Explanation:

QUESTION NO: 39

Which statement is a benefit of using Cisco IOS IPS?

A. It uses the underlying routing infrastructure to provide an additional layer of security.
B. It works in passive mode so as not to impact traffic flow.
C. It supports the complete signature database as a Cisco IPS sensor appliance.
D. The signature database is tied closely with the Cisco IOS image.

Answer: A
Explanation:

QUESTION NO: 40

You are the security administrator for a large enterprise network with many remote locations. You have been given the assignment to deploy a Cisco IPS solution.

Where in the network would be the best place to deploy Cisco IOS IPS?

A. inside the firewall of the corporate headquarters Internet connection
B. at the entry point into the data center
C. outside the firewall of the corporate headquarters Internet connection
D. at remote branch offices

Answer: D
Explanation:

QUESTION NO: 41

Which IPS technique commonly is used to improve accuracy and context awareness, aiming to detect and respond to relevant incidents only and therefore, reduce noise?

A. attack relevancy
B. target asset value
C. signature accuracy
D. risk rating

Answer: D
Explanation:

QUESTION NO: 42

Which two statements about SSL-based VPNs are true? (Choose two.)

A. Asymmetric algorithms are used for authentication and key exchange.
B. SSL VPNs and IPsec VPNs cannot be configured concurrently on the same router.
C. The application programming interface can be used to modify extensively the SSL client software for use in special applications.
D. The authentication process uses hashing technologies.
E. Both client and clientless SSL VPNs require special-purpose client software to be installed on the client machine.

Answer: A,D
Explanation:

QUESTION NO: 43

Which option describes the purpose of Diffie-Hellman?

A. used between the initiator and the responder to establish a basic security policy
B. used to verify the identity of the peer
C. used for asymmetric public key encryption
D. used to establish a symmetric shared key via a public key exchange process

Answer: D
Explanation:

QUESTION NO: 44

Which three statements about the IPsec ESP modes of operation are true? (Choose three.)

A. Tunnel mode is used between a host and a security gateway.
B. Tunnel mode is used between two security gateways.
C. Tunnel mode only encrypts and authenticates the data.
D. Transport mode authenticates the IP header.
E. Transport mode leaves the original IP header in the clear.

Answer: A,B,E
Explanation:

QUESTION NO: 45

When configuring SSL VPN on the Cisco ASA appliance, which configuration step is required only for Cisco AnyConnect full tunnel SSL VPN access and not required for clientless SSL VPN?

A. user authentication
B. group policy
C. IP address pool
D. SSL VPN interface
E. connection profile

Answer: C
Explanation:

QUESTION NO: 46

For what purpose is the Cisco ASA appliance web launch SSL VPN feature used?

A. to enable split tunneling when using clientless SSL VPN access
B. to enable users to login to a web portal to download and launch the AnyConnect client
C. to enable smart tunnel access for applications that are not web-based
D. to optimize the SSL VPN connections using DTLS
E. to enable single-sign-on so the SSL VPN users need only log in once

Answer: B
Explanation:

QUESTION NO: 47

Which statement describes how VPN traffic is encrypted to provide confidentiality when using asymmetric encryption?

A. The sender encrypts the data using the sender's private key, and the receiver decrypts the data using the sender's public key.
B. The sender encrypts the data using the sender's public key, and the receiver decrypts the data using the sender's private key.
C. The sender encrypts the data using the sender's public key, and the receiver decrypts the data using the receiver's public key.
D. The sender encrypts the data using the receiver's private key, and the receiver decrypts the data using the receiver's public key.
E. The sender encrypts the data using the receiver's public key, and the receiver decrypts the data using the receiver's private key.
F. The sender encrypts the data using the receiver's private key, and the receiver decrypts the data using the sender's public key.

Answer: E
Explanation:

QUESTION NO: 48

Which four types of VPN are supported using Cisco ISRs and Cisco ASA appliances? (Choose four.)

A. SSL clientless remote-access VPNs
B. SSL full-tunnel client remote-access VPNs
C. SSL site-to-site VPNs
D. IPsec site-to-site VPNs
E. IPsec client remote-access VPNs
F. IPsec clientless remote-access VPNs

Answer: A,B,D,E
Explanation:

QUESTION NO: 49

Which description of the Diffie-Hellman protocol is true?

A. It uses symmetrical encryption to provide data confidentiality over an unsecured communications channel.
B. It uses asymmetrical encryption to provide authentication over an unsecured communications

channel.
C. It is used within the IKE Phase 1 exchange to provide peer authentication.
D. It provides a way for two peers to establish a shared-secret key, which only they will know, even though they are communicating over an unsecured channel.
E. It is a data integrity algorithm that is used within the IKE exchanges to guarantee the integrity of the message of the IKE exchanges.

Answer: D
Explanation:

QUESTION NO: 50

Which IPsec transform set provides the strongest protection?

A. crypto ipsec transform-set 1 esp-3des esp-sha-hmac
B. crypto ipsec transform-set 2 esp-3des esp-md5-hmac
C. crypto ipsec transform-set 3 esp-aes 256 esp-sha-hmac
D. crypto ipsec transform-set 4 esp-aes esp-md5-hmac
E. crypto ipsec transform-set 5 esp-des esp-sha-hmac
F. crypto ipsec transform-set 6 esp-des esp-md5-hmac

Answer: C
Explanation:

QUESTION NO: 51

Which two options are characteristics of the Cisco Configuration Professional Security Audit wizard? (Choose two.)

A. displays a screen with fix-it check boxes to let you choose which potential security-related configuration changes to implement
B. has two modes of operation: interactive and non-interactive
C. automatically enables Cisco IOS firewall and Cisco IOS IPS to secure the router
D. uses interactive dialogs and prompts to implement role-based CLI
E. requires users to first identify which router interfaces connect to the inside network and which connect to the outside network

Answer: A,E
Explanation:

QUESTION NO: 52

Which statement describes a result of securing the Cisco IOS image using the Cisco IOS image resilience feature?

A. The show version command does not show the Cisco IOS image file location.
B. The Cisco IOS image file is not visible in the output from the show flash command.
C. When the router boots up, the Cisco IOS image is loaded from a secured FTP location.
D. The running Cisco IOS image is encrypted and then automatically backed up to the NVRAM.
E. The running Cisco IOS image is encrypted and then automatically backed up to a TFTP server.

Answer: B
Explanation:

QUESTION NO: 53

Which aaa accounting command is used to enable logging of the start and stop records for user terminal sessions on the router?

A. aaa accounting network start-stop tacacs+
B. aaa accounting system start-stop tacacs+
C. aaa accounting exec start-stop tacacs+
D. aaa accounting connection start-stop tacacs+
E. aaa accounting commands 15 start-stop tacacs+

Answer: C
Explanation:

QUESTION NO: 54

Which access list permits HTTP traffic sourced from host 10.1.129.100 port 3030 destined to host 192.168.1.10?

A. access-list 101 permit tcp any eq 3030
B. access-list 101 permit tcp 10.1.128.0 0.0.1.255 eq 3030 192.168.1.0 0.0.0.15 eq www
C. access-list 101 permit tcp 10.1.129.0 0.0.0.255 eq www 192.168.1.10 0.0.0.0 eq www
D. access-list 101 permit tcp host 192.168.1.10 eq 80 10.1.0.0 0.0.255.255 eq 3030
E. access-list 101 permit tcp 192.168.1.10 0.0.0.0 eq 80 10.1.0.0 0.0.255.255
F. access-list 101 permit ip host 10.1.129.100 eq 3030 host 192.168.1.100 eq 80

Answer: B

Explanation:

QUESTION NO: 55

Which location is recommended for extended or extended named ACLs?

A. an intermediate location to filter as much traffic as possible
B. a location as close to the destination traffic as possible
C. when using the established keyword, a location close to the destination point to ensure that return traffic is allowed
D. a location as close to the source traffic as possible

Answer: D
Explanation:

QUESTION NO: 56

Which statement about asymmetric encryption algorithms is true?

A. They use the same key for encryption and decryption of data.
B. They use the same key for decryption but different keys for encryption of data.
C. They use different keys for encryption and decryption of data.
D. They use different keys for decryption but the same key for encryption of data.

Answer: C
Explanation:

QUESTION NO: 57

Which option can be used to authenticate the IPsec peers during IKE Phase 1?

A. Diffie-Hellman Nonce
B. pre-shared key
C. XAUTH
D. integrity check value
E. ACS
F. AH

Answer: B
Explanation:

QUESTION NO: 58

Which single Cisco IOS ACL entry permits IP addresses from 172.16.80.0 to 172.16.87.255?

A. permit 172.16.80.0 0.0.3.255
B. permit 172.16.80.0 0.0.7.255
C. permit 172.16.80.0 0.0.248.255
D. permit 176.16.80.0 255.255.252.0
E. permit 172.16.80.0 255.255.248.0
F. permit 172.16.80.0 255.255.240.0

Answer: B
Explanation:

QUESTION NO: 59

You want to use the Cisco Configuration Professional site-to-site VPN wizard to implement a site-to-site IPsec VPN using pre-shared key.

Which four configurations are required (with no defaults)? (Choose four.)

A. the interface for the VPN connection
B. the VPN peer IP address
C. the IPsec transform-set
D. the IKE policy
E. the interesting traffic (the traffic to be protected)
F. the pre-shared key

Answer: A,B,E,F
Explanation:

QUESTION NO: 60

Which two options represent a threat to the physical installation of an enterprise network? (Choose two.)

A. surveillance camera
B. security guards
C. electrical power
D. computer room access
E. change control

Answer: C,D
Explanation:

QUESTION NO: 61

Which option represents a step that should be taken when a security policy is developed?

A. Perform penetration testing.
B. Determine device risk scores.
C. Implement a security monitoring system.
D. Perform quantitative risk analysis.

Answer: D
Explanation:

QUESTION NO: 62

Which type of network masking is used when Cisco IOS access control lists are configured?

A. extended subnet masking
B. standard subnet masking
C. priority masking
D. wildcard masking

Answer: D
Explanation:

QUESTION NO: 63

How are Cisco IOS access control lists processed?

A. Standard ACLs are processed first.

B. The best match ACL is matched first.
C. Permit ACL entries are matched first before the deny ACL entries.
D. ACLs are matched from top down.
E. The global ACL is matched first before the interface ACL.

Answer: D
Explanation:

QUESTION NO: 64

Which type of management reporting is defined by separating management traffic from production traffic?

A. IPsec encrypted
B. in-band
C. out-of-band
D. SSH

Answer: C
Explanation:

QUESTION NO: 65

Which syslog level is associated with LOG_WARNING?

A. 1
B. 2
C. 3
D. 4
E. 5
F. 6

Answer: D
Explanation:

QUESTION NO: 66

In which type of Layer 2 attack does an attacker broadcast BDPUs with a lower switch priority?

A. MAC spoofing attack
B. CAM overflow attack
C. VLAN hopping attack
D. STP attack

Answer: D
Explanation:

QUESTION NO: 67

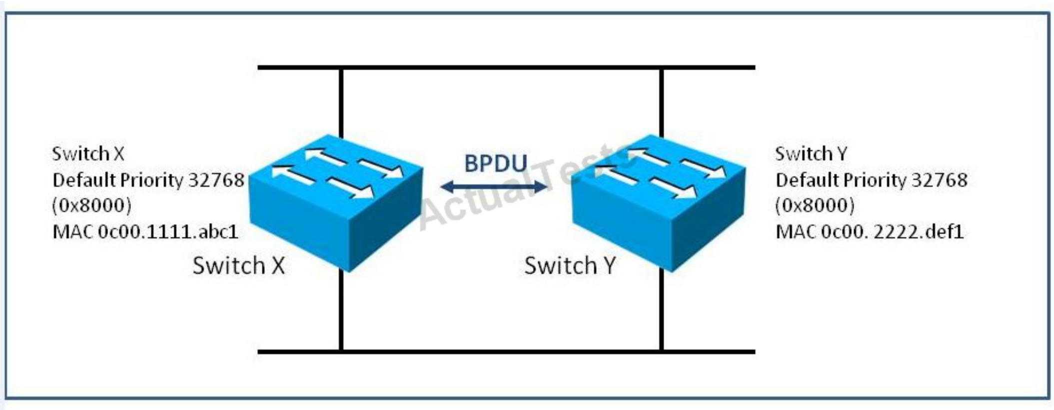
Which security measure must you take for native VLANs on a trunk port?

A. Native VLANs for trunk ports should never be used anywhere else on the switch.
B. The native VLAN for trunk ports should be VLAN 1.
C. Native VLANs for trunk ports should match access VLANs to ensure that cross-VLAN traffic from multiple switches can be delivered to physically disparate switches.
D. Native VLANs for trunk ports should be tagged with 802.1Q.

Answer: A
Explanation:

QUESTION NO: 68

Refer to the exhibit.



Which switch is designated as the root bridge in this topology?

A. It depends on which switch came on line first.
B. Neither switch would assume the role of root bridge because they have the same default priority.
C. switch X
D. switch Y

Answer: C
Explanation:

QUESTION NO: 69

Which type of firewall technology is considered the versatile and commonly used firewall technology?

A. static packet filter firewall
B. application layer firewall
C. stateful packet filter firewall
D. proxy firewall
E. adaptive layer firewall

Answer: C
Explanation:

QUESTION NO: 70

Which type of NAT is used where you translate multiple internal IP addresses to a single global, routable IP address?

A. policy NAT
B. dynamic PAT
C. static NAT
D. dynamic NAT
E. policy PAT

Answer: B
Explanation:

QUESTION NO: 71

Which Cisco IPS product offers an inline, deep-packet inspection feature that is available in integrated services routers?

A. Cisco iSDM
B. Cisco AIM
C. Cisco IOS IPS
D. Cisco AIP-SSM

Answer: C
Explanation:

QUESTION NO: 72

Which three modes of access can be delivered by SSL VPN? (Choose three.)

A. full tunnel client
B. IPsec SSL
C. TLS transport mode
D. thin client
E. clientless
F. TLS tunnel mode

Answer: A,D,E
Explanation:

QUESTION NO: 73

During role-based CLI configuration, what must be enabled before any user views can be created?

A. multiple privilege levels
B. usernames and passwords
C. aaa new-model command
D. secret password for the root user
E. HTTP and/or HTTPS server
F. TACACS server group

Answer: C
Explanation:

QUESTION NO: 74

Which three statements about applying access control lists to a Cisco router are true? (Choose three.)

A. Place more specific ACL entries at the top of the ACL.
B. Place generic ACL entries at the top of the ACL to filter general traffic and thereby reduce "noise" on the network.
C. ACLs always search for the most specific entry before taking any filtering action.
D. Router-generated packets cannot be filtered by ACLs on the router.
E. If an access list is applied but it is not configured, all traffic passes.

Answer: A,D,E
Explanation:

QUESTION NO: 75

When port security is enabled on a Cisco Catalyst switch, what is the default action when the configured maximum number of allowed MAC addresses value is exceeded?

A. The port remains enabled, but bandwidth is throttled until old MAC addresses are aged out.
B. The port is shut down.
C. The MAC address table is cleared and the new MAC address is entered into the table.
D. The violation mode of the port is set to restrict.

Answer: B
Explanation:

QUESTION NO: 76

Which three statements about the Cisco ASA appliance are true? (Choose three.)

A. The DMZ interface(s) on the Cisco ASA appliance most typically use a security level between 1 and 99.
B. The Cisco ASA appliance supports Active/Active or Active/Standby failover.
C. The Cisco ASA appliance has no default MPF configurations.
D. The Cisco ASA appliance uses security contexts to virtually partition the ASA into multiple virtual firewalls.
E. The Cisco ASA appliance supports user-based access control using 802.1x.
F. An SSM is required on the Cisco ASA appliance to support Botnet Traffic Filtering.

Answer: A,B,D

Explanation:

QUESTION NO: 77

Refer to the exhibit.

```
access-list 102 permit tcp any 192.168.15.32 0.0.0.31 eq www
access-list 102 deny ip any 192.168.15.32 0.0.0.31
access-list 102 permit ip any any
```

```
Current configuration : 156 bytes
!
interface FastEthernet0
 ip address 192.168.32.13 255.255.255.0
 ip access-group IOS in
 ip flow ingress
 ip flow egress
 duplex auto
 speed auto
 !
end

Router(config-ext-nacl)#do show access-l IOS
Extended IP access list IOS
    10 permit tcp host 10.1.1.1 eq 1030 host 192.168.15.80 eq telnet
    20 permit tcp host 10.1.1.1 eq 1030 host 192.168.15.66 eq 8080
    30 permit tcp host 10.1.1.1 eq 1030 host 192.168.15.36 eq www
    40 permit tcp host 10.1.1.1 eq 1030 host 192.168.15.63 eq www
Router(config-ext-nacl)#
```

This Cisco IOS access list has been configured on the FA0/0 interface in the inbound direction.

Which four TCP packets sourced from 10.1.1.1 port 1030 and routed to the FA0/0 interface are permitted? (Choose four.)

A. destination ip address: 192.168.15.37 destination port: 22
B. destination ip address: 192.168.15.80 destination port: 23
C. destination ip address: 192.168.15.66 destination port: 8080
D. destination ip address: 192.168.15.36 destination port: 80
E. destination ip address: 192.168.15.63 destination port: 80
F. destination ip address: 192.168.15.40 destination port: 21

Answer: B,C,D,E

Explanation:

QUESTION NO: 78

You use Cisco Configuration Professional to enable Cisco IOS IPS. Which state must a signature be in before any actions can be taken when an attack matches that signature?

A. enabled
B. unretired
C. successfully complied
D. successfully complied and unretired
E. successfully complied and enabled
F. unretired and enabled
G. enabled, unretired, and successfully complied

Answer: G
Explanation:

QUESTION NO: 79

Which statement describes how the sender of the message is verified when asymmetric encryption is used?

A. The sender encrypts the message using the sender's public key, and the receiver decrypts the message using the sender's private key.
B. The sender encrypts the message using the sender's private key, and the receiver decrypts the message using the sender's public key.
C. The sender encrypts the message using the receiver's public key, and the receiver decrypts the message using the receiver's private key.
D. The sender encrypts the message using the receiver's private key, and the receiver decrypts the message using the receiver's public key.
E. The sender encrypts the message using the receiver's public key, and the receiver decrypts the message using the sender's public key.

Answer: B
Explanation:

QUESTION NO: 80

Refer to the exhibit.

```
router#show crypto isakmp policy
Protection suite of priority 1
   encryption algorithm: 3DES - Data Encryption Standard (168 bit keys).
   hash algorithm: Secure Hash Standard
   authentication method: preshared
   Diffie-Hellman group: #2 (1024 bit)
   lifetime: 86400 seconds, no volume limit
Protection suite of priority 2
   encryption algorithm: DES - Data Encryption Standard (56 bit keys).
   hash algorithm: Secure Hash Standard
   authentication method: preshared
   Diffie-Hellman group: #2 (1024 bit)
   lifetime: 86400 seconds, no volume limit
Default protection suite
   encryption algorithm: DES - Data Encryption Standard (56 bit keys).
   hash algorithm: Secure Hash Standard
   authentication method: Rivest-Shamir-Adleman Signature
   Diffie-Hellman group: #1 (768 bit)
   lifetime: 86400 seconds, no volume limit

router#show crypto ipsec transform-set
Transform set mine: { esp-128-aes esp-sha-hmac } will negotiate = { Tunnel, },

router#show crypto map
Crypto Map "mymap" 10 ipsec-isakmp
   Peer = 172.16.1.2
   Extended IP access list 110
      access-list 110 permit ip 10.10.10.0 0.0.0.255 10.10.20.0 0.0.0.255
   Current peer: 172.16.1.2
   Security association lifetime: 4608000 kilobytes/3600 seconds
   PFS (Y/N): N
   Transform sets={ mine, }
```

Which three statements about these three show outputs are true? (Choose three.)

A. Traffic matched by ACL 110 is encrypted.
B. The IPsec transform set uses SHA for data confidentiality.
C. The crypto map shown is for an IPsec site-to-site VPN tunnel.
D. The default ISAKMP policy uses a digital certificate to authenticate the IPsec peer.
E. The IPsec transform set specifies the use of GRE over IPsec tunnel mode.
F. The default ISAKMP policy has higher priority than the other two ISAKMP policies with a priority of 1 and 2

Answer: A,C,D
Explanation:

QUESTION NO: 81

Which type of security control is defense in depth?

A. threat mitigation
B. risk analysis
C. botnet mitigation
D. overt and covert channels

Answer: A
Explanation:

QUESTION NO: 82

Which two options are two of the built-in features of IPv6? (Choose two.)

A. VLSM
B. native IPsec
C. controlled broadcasts
D. mobile IP
E. NAT

Answer: B,D
Explanation:

QUESTION NO: 83

Which option is a characteristic of the RADIUS protocol?

A. uses TCP
B. offers multiprotocol support
C. combines authentication and authorization in one process
D. supports bi-directional challenge

Answer: C
Explanation:

QUESTION NO: 84

Refer to the below.

```
Router# debug tacacs

14:00:09: TAC+: Opening TCP/IP connection to 192.168.60.15 using source
10.116.0.79
14:00:09: TAC+: Sending TCP/IP packet number 383258052-1 to 192.168.60.15
(AUTHEN/START)
14:00:09: TAC+: Receiving TCP/IP packet number 383258052-2 from 192.168.60.15
14:00:09: TAC+ (383258052): received authen response status = GETUSER
14:00:10: TAC+: send AUTHEN/CONT packet
14:00:10: TAC+: Sending TCP/IP packet number 383258052-3 to 192.168.60.15
(AUTHEN/CONT)
14:00:10: TAC+: Receiving TCP/IP packet number 383258052-4 from 192.168.60.15
14:00:10: TAC+ (383258052): received authen response status = GETPASS
14:00:14: TAC+: send AUTHEN/CONT packet
14:00:14: TAC+: Sending TCP/IP packet number 383258052-5 to 192.168.60.15
(AUTHEN/CONT)
14:00:14: TAC+: Receiving TCP/IP packet number 383258052-6 from 192.168.60.15
14:00:14: TAC+ (383258052): received authen response status = PASS
14:00:14: TAC+: Closing TCP/IP connection to 192.168.60.15
```


Which statement about this debug output is true?

A. The requesting authentication request came from username GETUSER.
B. The TACACS+ authentication request came from a valid user.
C. The TACACS+ authentication request passed, but for some reason the user's connection was closed immediately.
D. The initiating connection request was being spoofed by a different source address.

Answer: B
Explanation:

QUESTION NO: 85

Which type of Cisco IOS access control list is identified by 100 to 199 and 2000 to 2699?

A. standard
B. extended
C. named
D. IPv4 for 100 to 199 and IPv6 for 2000 to 2699

Answer: B

Explanation:

QUESTION NO: 86

Which priority is most important when you plan out access control lists?

A. Build ACLs based upon your security policy.
B. Always put the ACL closest to the source of origination.
C. Place deny statements near the top of the ACL to prevent unwanted traffic from passing through the router.
D. Always test ACLs in a small, controlled production environment before you roll it out into the larger production network.

Answer: A

Explanation:

QUESTION NO: 87

Which step is important to take when implementing secure network management?

A. Implement in-band management whenever possible.
B. Implement telnet for encrypted device management access.
C. Implement SNMP with read/write access for troubleshooting purposes.
D. Synchronize clocks on hosts and devices.
E. Implement management plane protection using routing protocol authentication.

Answer: D

Explanation:

QUESTION NO: 88

Which statement best represents the characteristics of a VLAN?

A. Ports in a VLAN will not share broadcasts amongst physically separate switches.
B. A VLAN can only connect across a LAN within the same building.
C. A VLAN is a logical broadcast domain that can span multiple physical LAN segments.
D. A VLAN provides individual port security.

Answer: C

Explanation:

QUESTION NO: 89

Which Layer 2 protocol provides loop resolution by managing the physical paths to given network segments?

A. root guard
B. port fast
C. HSRP
D. STP

Answer: D

Explanation:

QUESTION NO: 90

When STP mitigation features are configured, where should the root guard feature be deployed?

A. toward ports that connect to switches that should not be the root bridge
B. on all switch ports
C. toward user-facing ports
D. Root guard should be configured globally on the switch.

Answer: A

Explanation:

QUESTION NO: 91

Which option is a characteristic of a stateful firewall?

A. can analyze traffic at the application layer
B. allows modification of security rule sets in real time to allow return traffic
C. will allow outbound communication, but return traffic must be explicitly permitted
D. supports user authentication

Answer: B

Explanation:

QUESTION NO: 92

Which type of NAT would you configure if a host on the external network required access to an internal host?

A. outside global NAT
B. NAT overload
C. dynamic outside NAT
D. static NAT

Answer: D
Explanation:

QUESTION NO: 93

Which statement about disabled signatures when using Cisco IOS IPS is true?

A. They do not take any actions, but do produce alerts.
B. They are not scanned or processed.
C. They still consume router resources.
D. They are considered to be "retired" signatures.

Answer: C
Explanation:

QUESTION NO: 94

Which type of intrusion prevention technology is the primary type used by the Cisco IPS security appliances?

A. profile-based
B. rule-based
C. protocol analysis-based
D. signature-based
E. NetFlow anomaly-based

Answer: D

Explanation:

QUESTION NO: 95

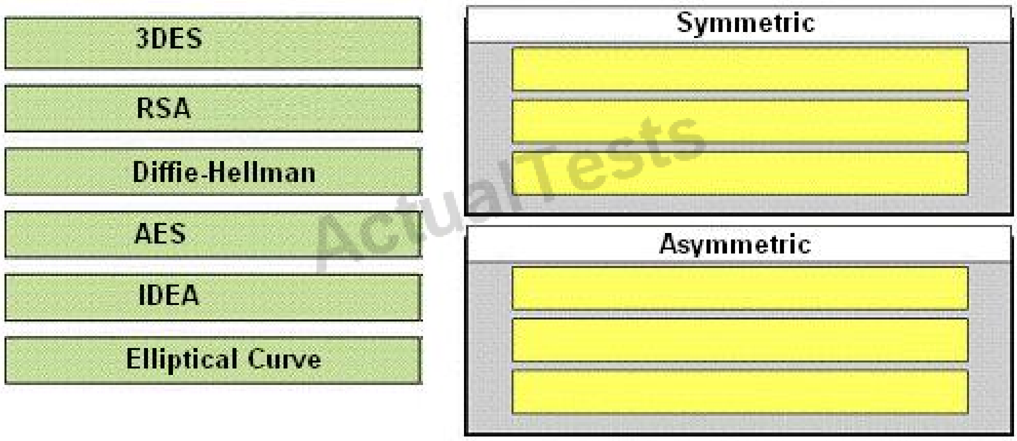
Which two services are provided by IPsec? (Choose two.)

A. Confidentiality
B. Encapsulating Security Payload
C. Data Integrity
D. Authentication Header
E. Internet Key Exchange

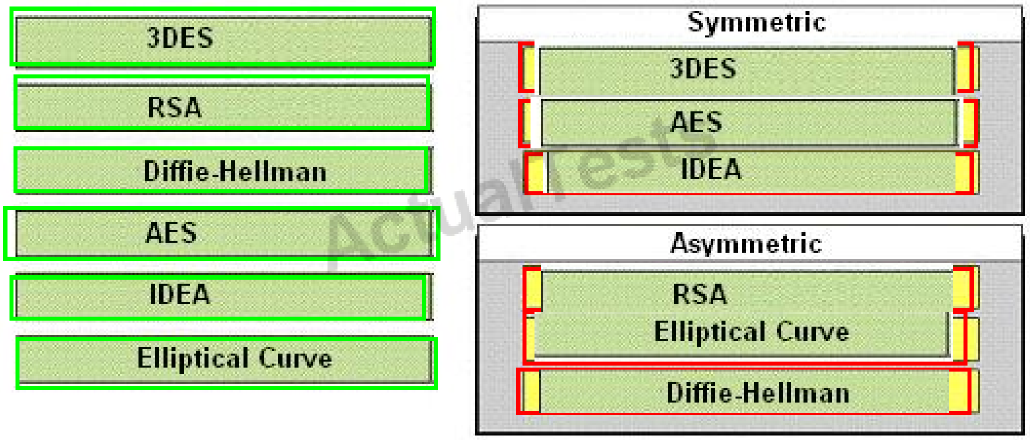
Answer: A,C

Explanation:

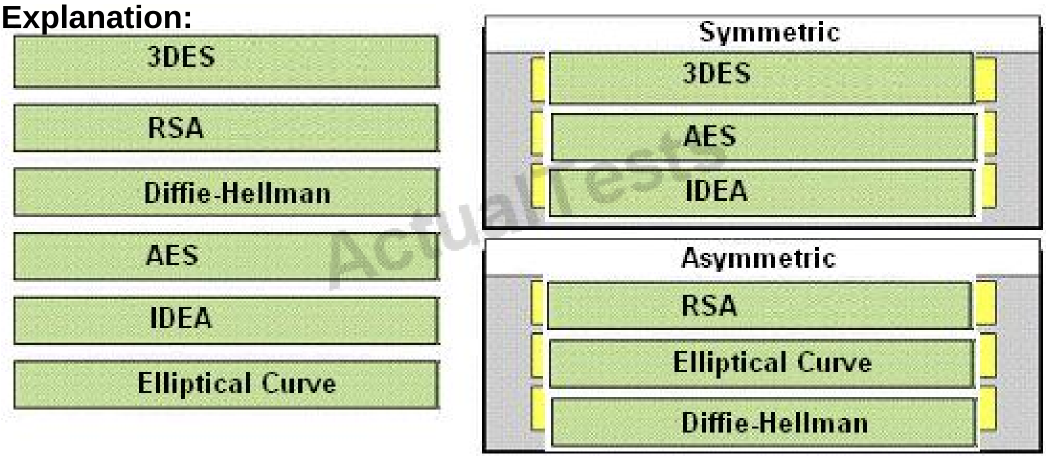
QUESTION NO: 96 DRAG DROP



Answer:



Explanation:



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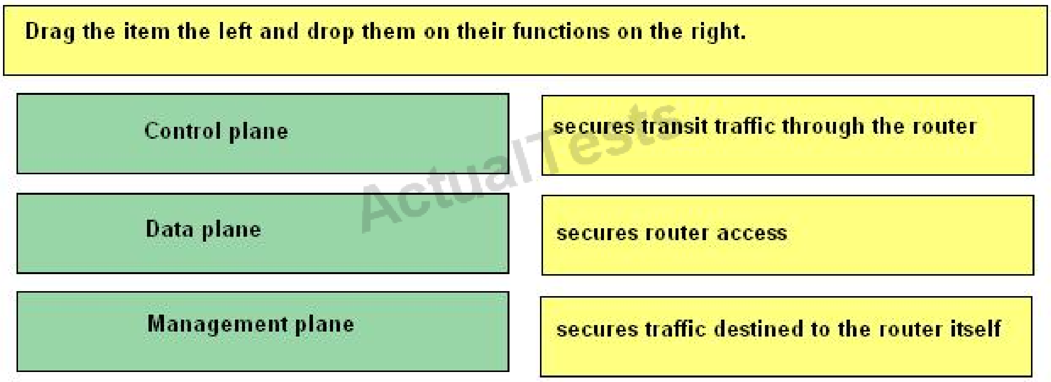
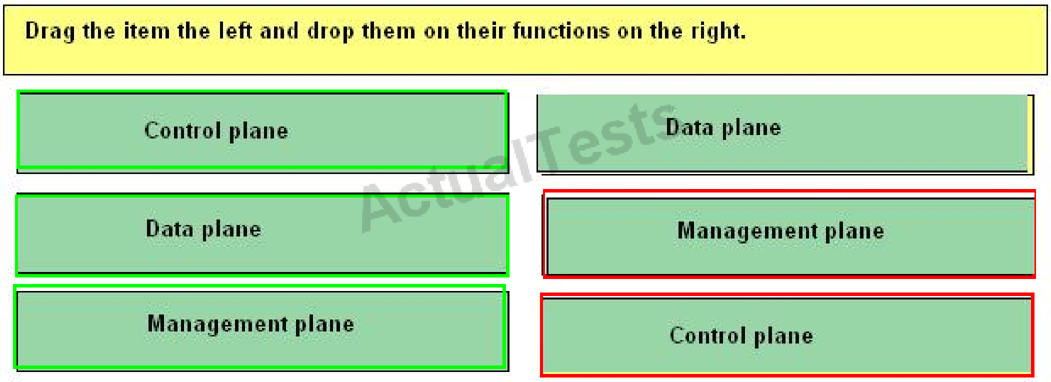
QUESTION NO: 97 DRAG DROP



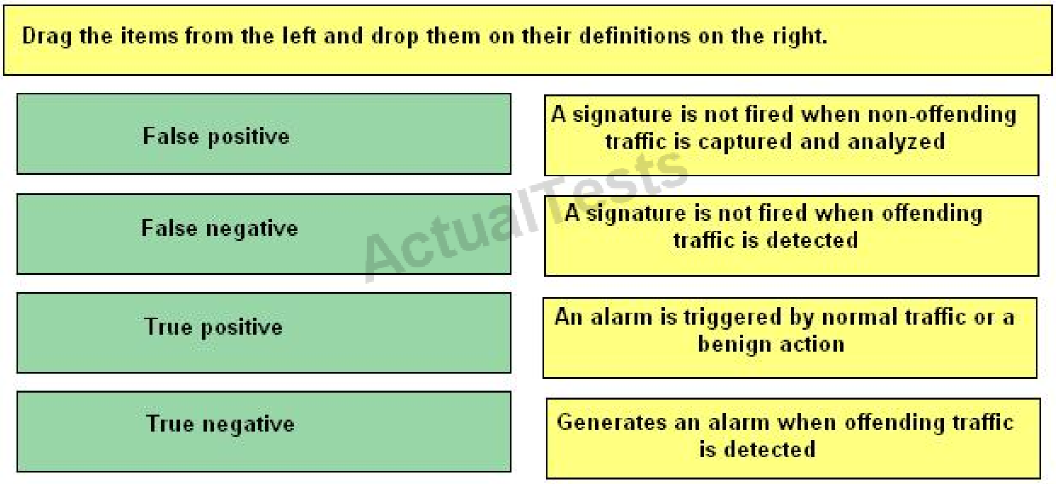
Answer:



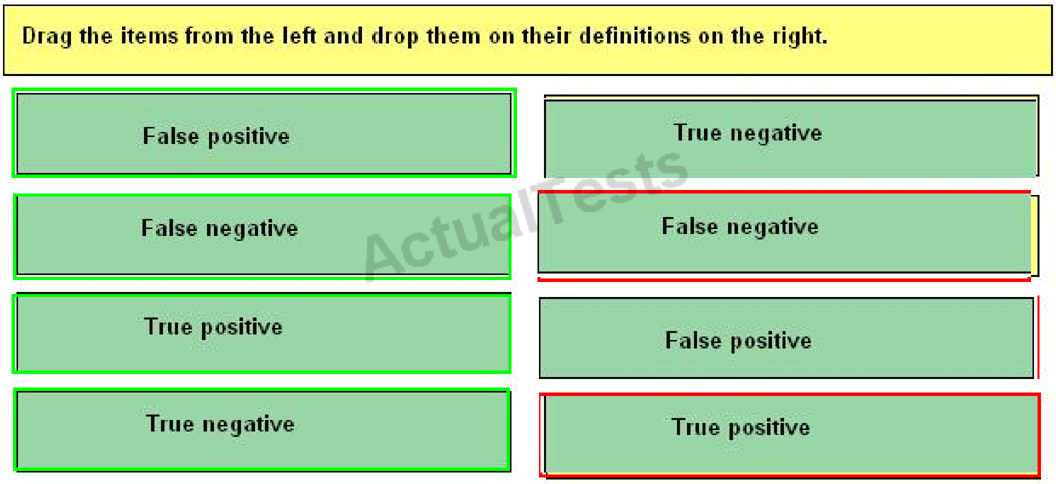
Explanation:

Control plane – secures traffic destined to the router itself
Data plane – secures transit traffic through the router
Management plane – secures router access

QUESTION NO: 98 DRAG DROP



Answer:



Explanation:

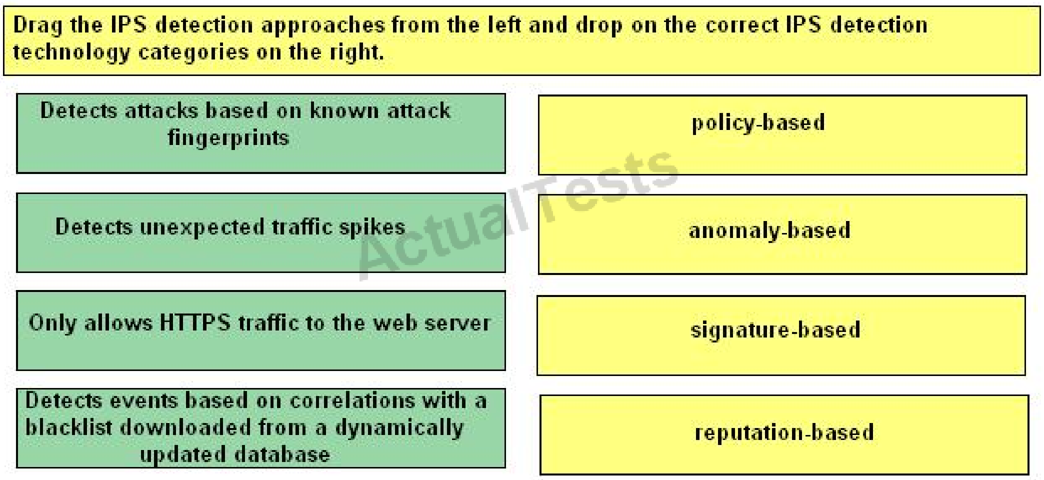
False positive – an alarm is triggered by normal traffic or a benign action

False negative – a signature is not fired when offending traffic is detected

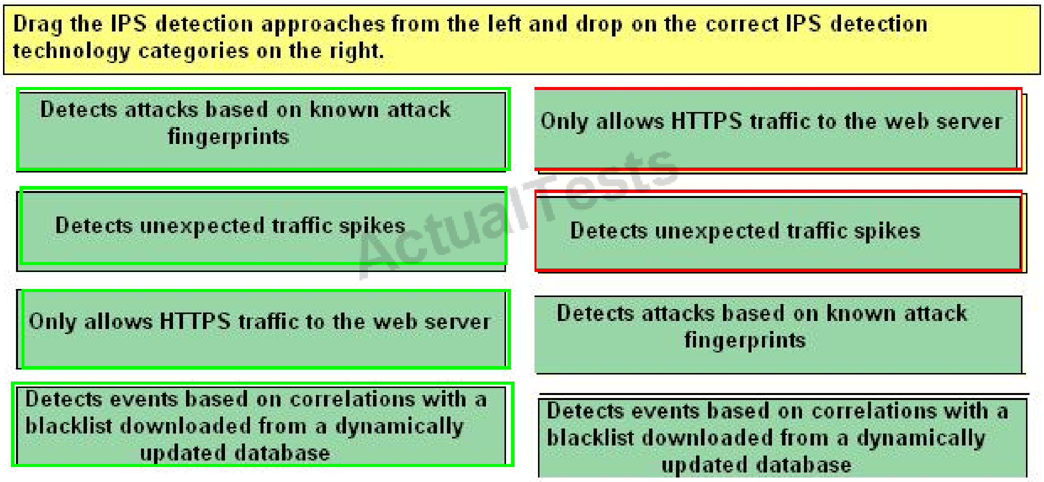
True positive – generates an alarm when offending traffic is detected

True negative – a signature is not fired when non-offending traffic is captured and analyzed

QUESTION NO: 99 DRAG DROP



Answer:



Explanation:

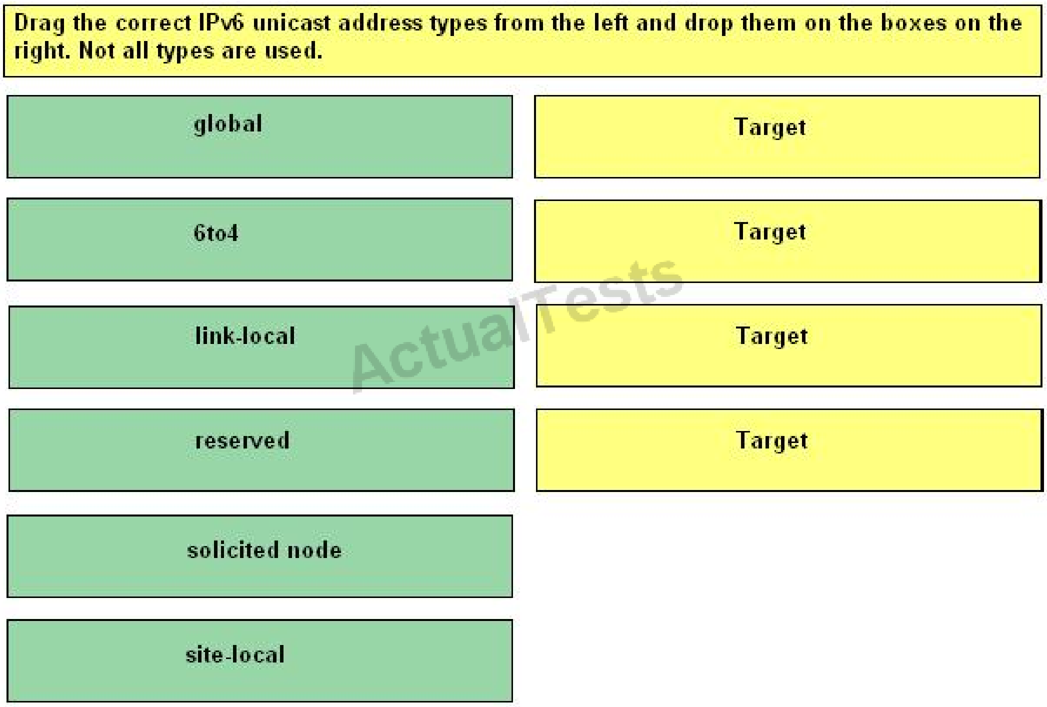
Detects attacks based on known attack fingerprints – signature-based

Detects unexpected traffic spikes – anomaly-based

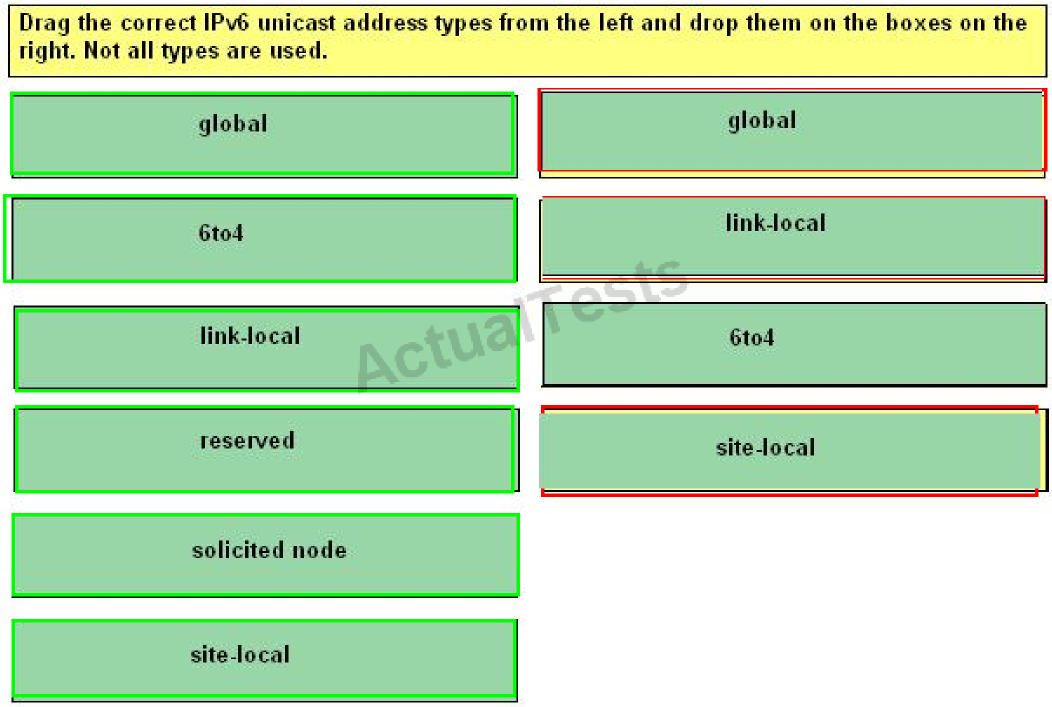
Only allows HTTPS traffic to the web server – policy-based

Detects events based on correlations with a blacklist downloaded from a dynamically updated database – reputation-based

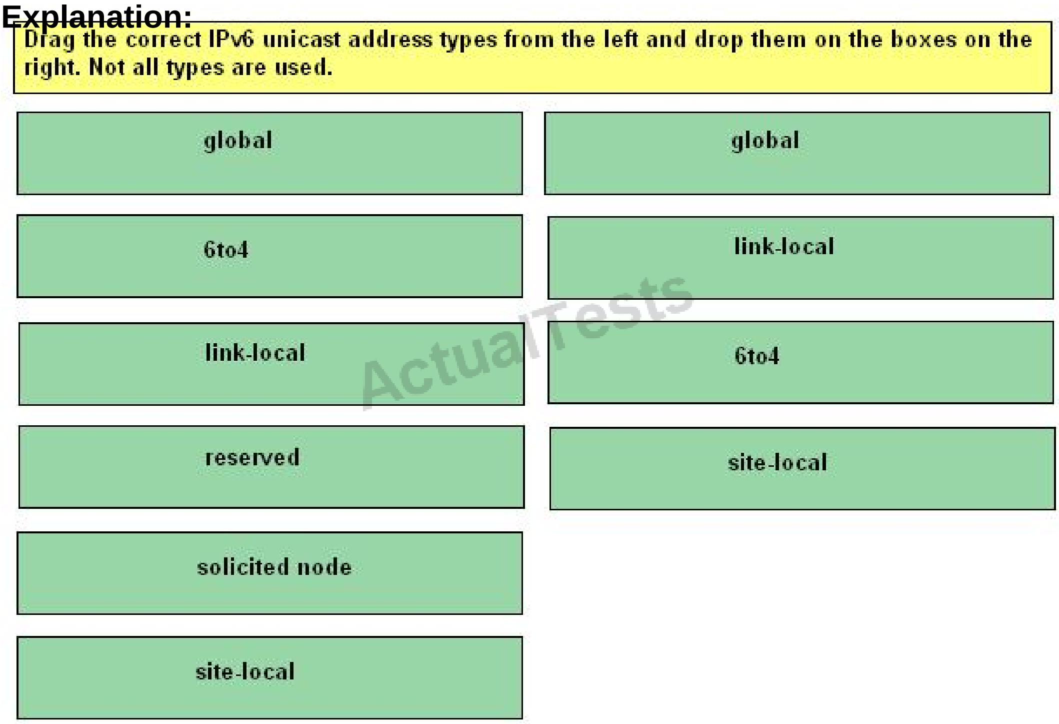
QUESTION NO: 100 DRAG DROP



Answer:



Explanation:



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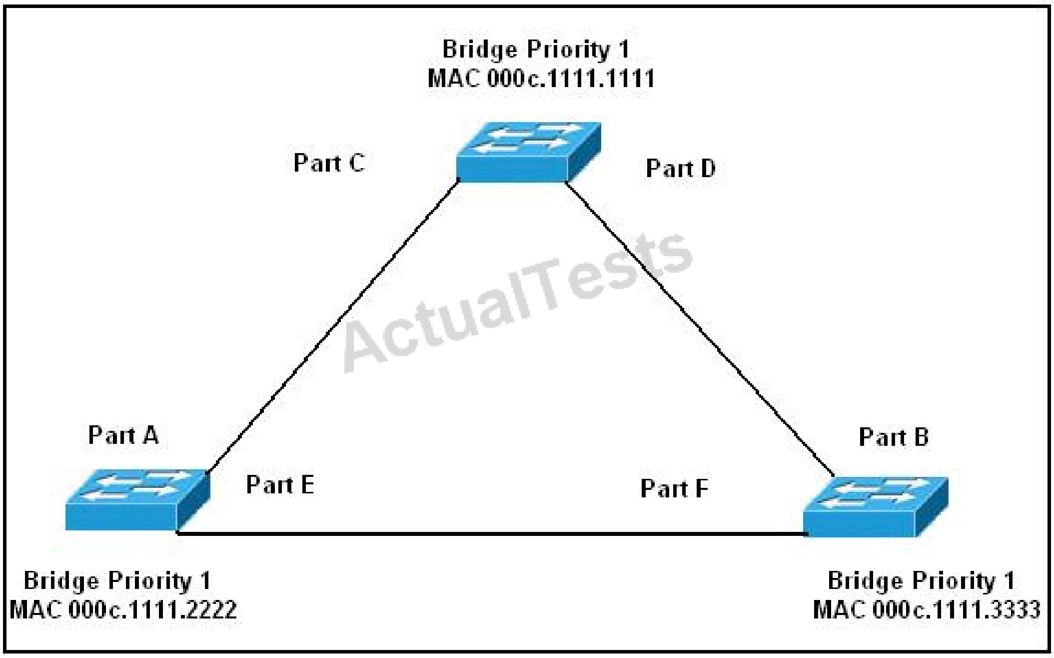
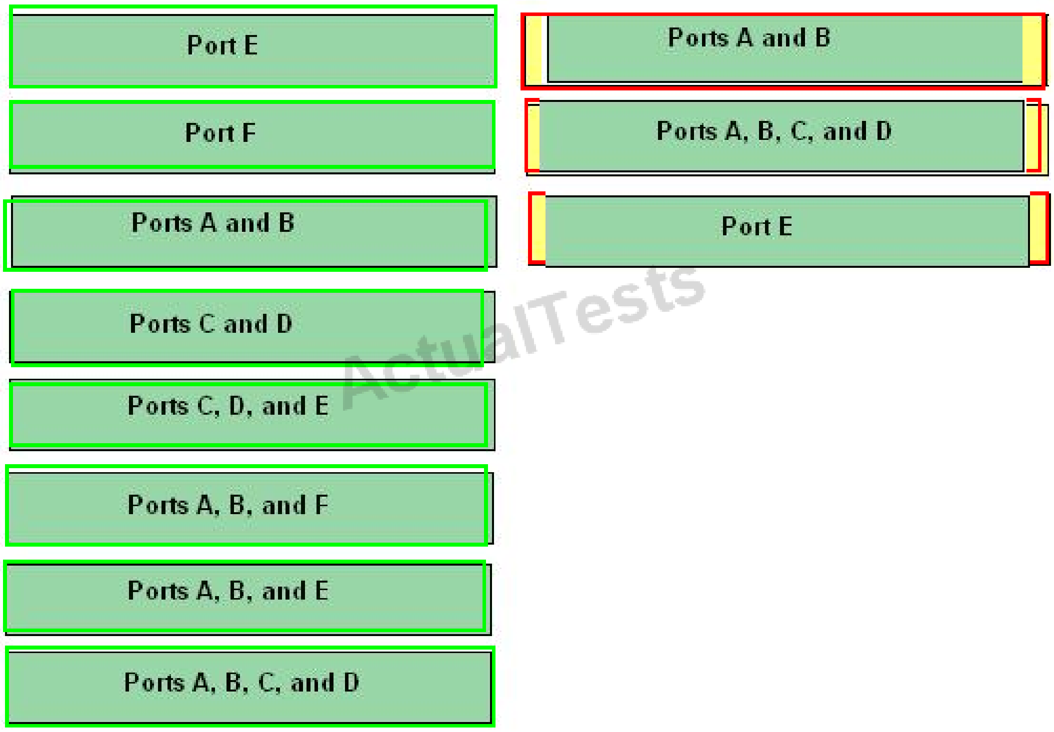
QUESTION NO: 101 DRAG DROP

Refer to the exhibit. Drag the port(s) from the left and drop them on the correct STP roles on the right. Not all options on the left are used.



Port E	root port(s)
Port F	designated port(s)
Ports A and B	non-designated port(s)
Ports C and D	
Ports C, D, and E	
Ports A, B, and F	
Ports A, B, and E	
Ports A, B, C, and D	

Answer:



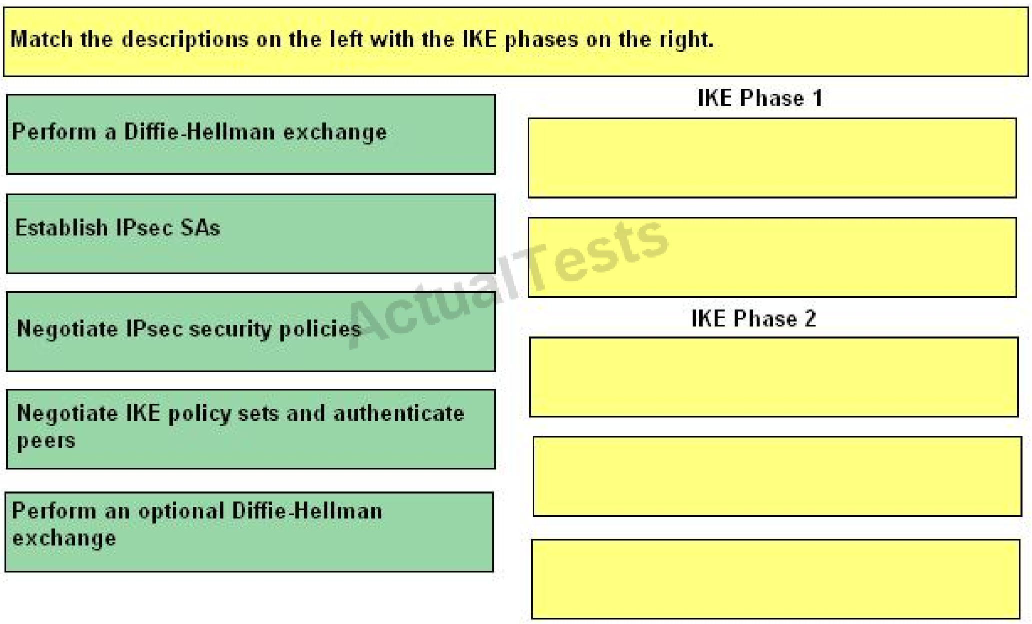
Explanation:

Ports A and B – root port(s)

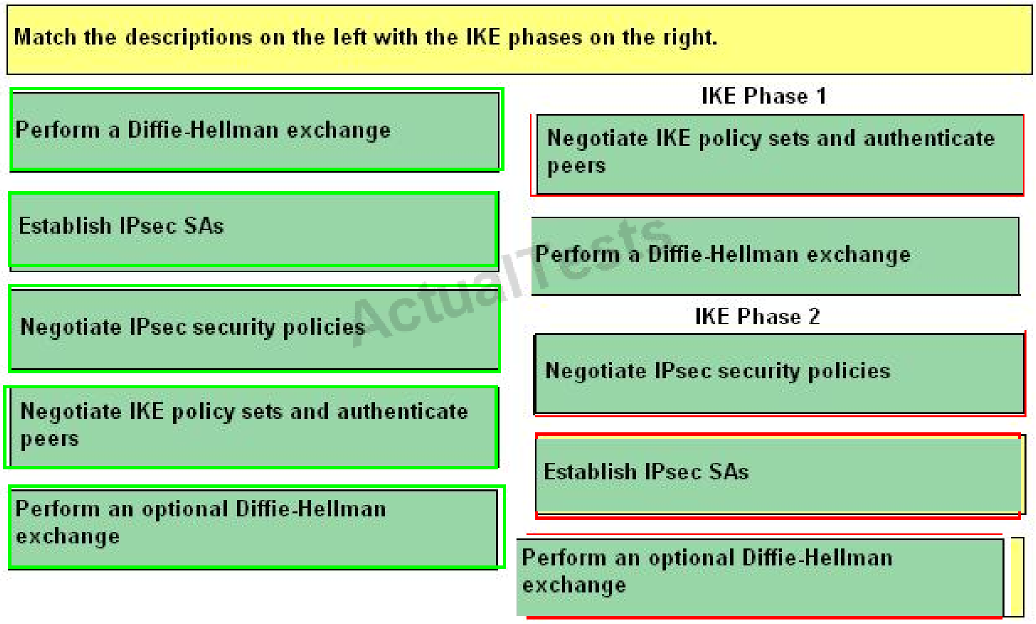
Ports A, B, C and D – designated port(s)

Ports E and F – non-designated port(s)

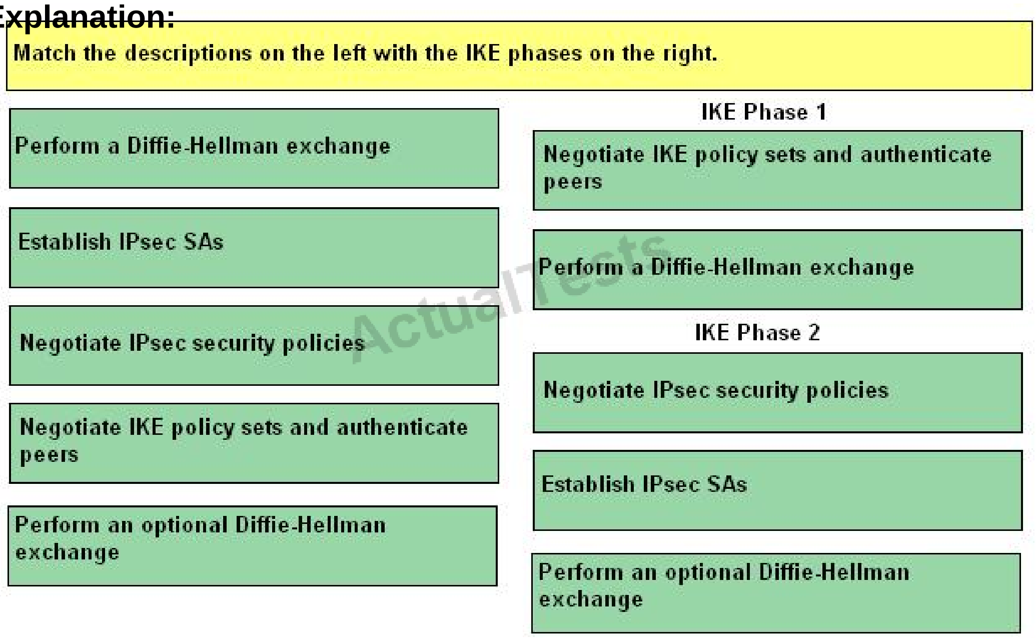
QUESTION NO: 102 DRAG DROP



Answer:



Explanation:



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QUESTION NO: 103 DRAG DROP

Drag the characteristics from the left and drop them the correct categories on the right.

- Can stop the attack trigger packet
- No network impact if there is a sensor overload
- Allows malicious traffic to pass before it can respond
- Deployed in promiscuous mode
- Can use stream normalization techniques
- More vulnerable to network evasion techniques
- Has some impact on network latency and jitter
- Deployed in inline mode

IPS

- Target
- Target
- Target
- Target

IDS

- Target
- Target
- Target
- Target

Answer:

Drag the characteristics from the left and drop them the correct categories on the right.

- Can stop the attack trigger packet
- No network impact if there is a sensor overload
- Allows malicious traffic to pass before it can respond
- Deployed in promiscuous mode
- Can use stream normalization techniques
- More vulnerable to network evasion techniques
- Has some impact on network latency and jitter
- Deployed in inline mode

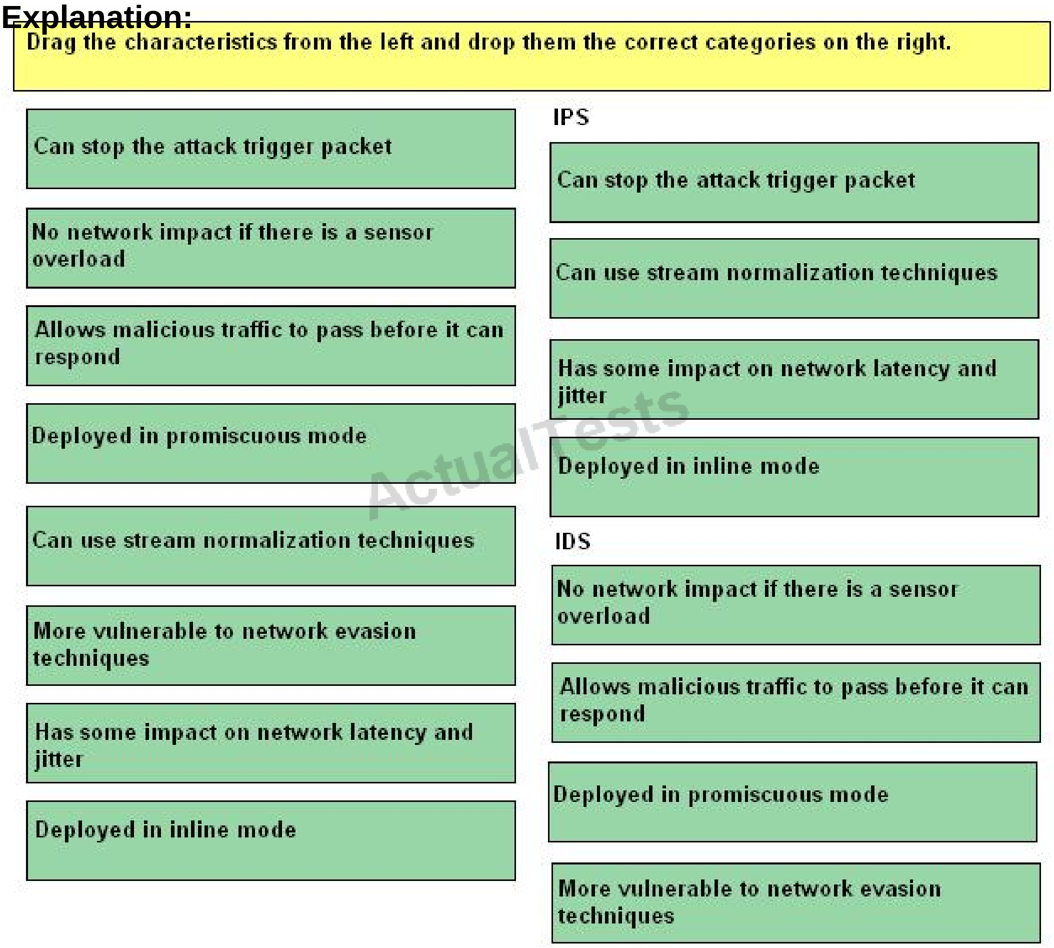
IPS

- Can stop the attack trigger packet
- Can use stream normalization techniques
- Has some impact on network latency and jitter
- Deployed in inline mode

IDS

- No network impact if there is a sensor overload
- Allows malicious traffic to pass before it can respond
- Deployed in promiscuous mode
- More vulnerable to network evasion techniques

Explanation:



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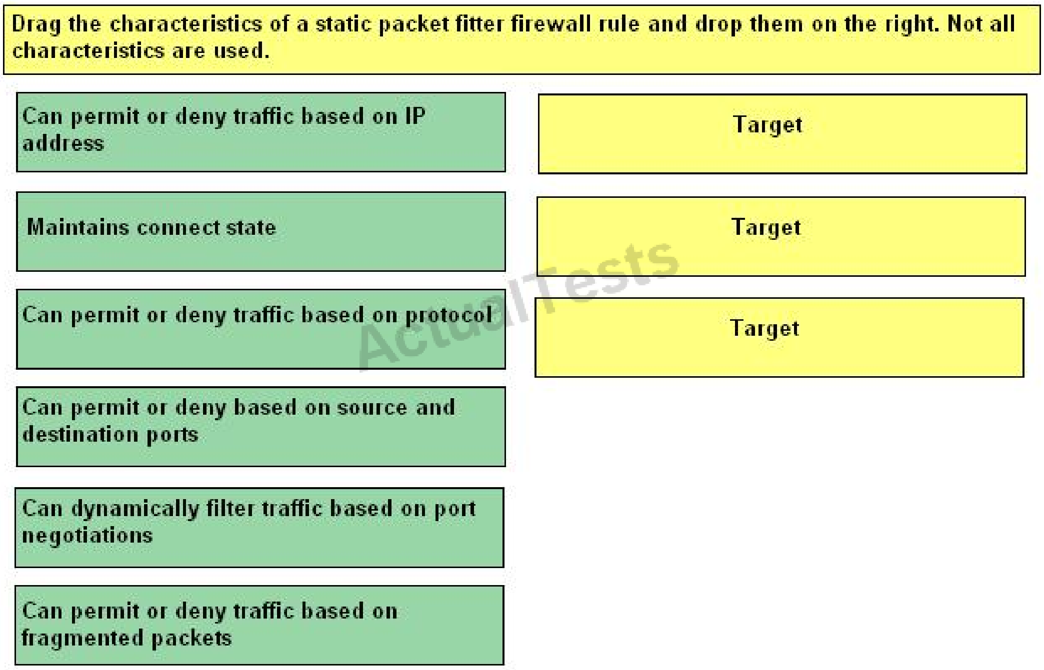
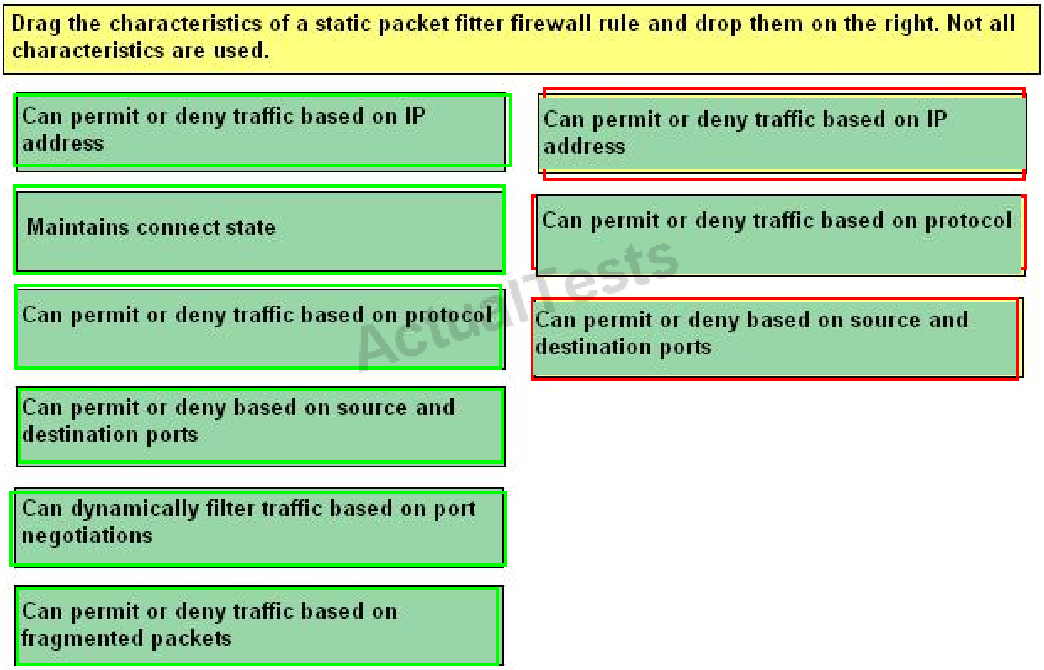
QUESTION NO: 104 DRAG DROP

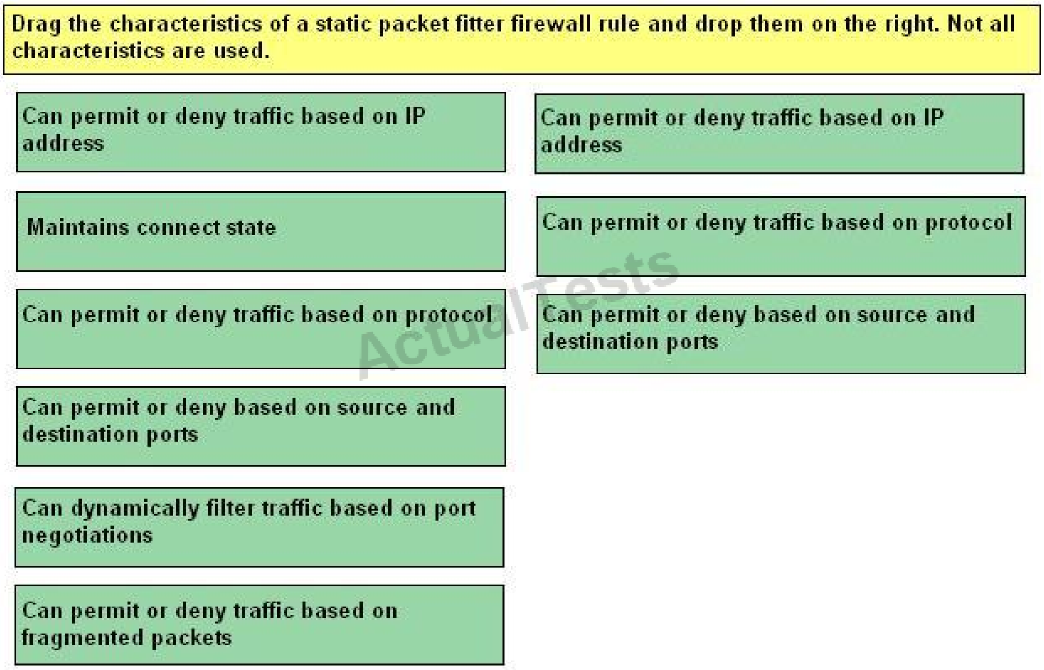
Drag the characteristics of a static packet fitter firewall rule and drop them on the right. Not all characteristics are used.



Answer:



Explanation:



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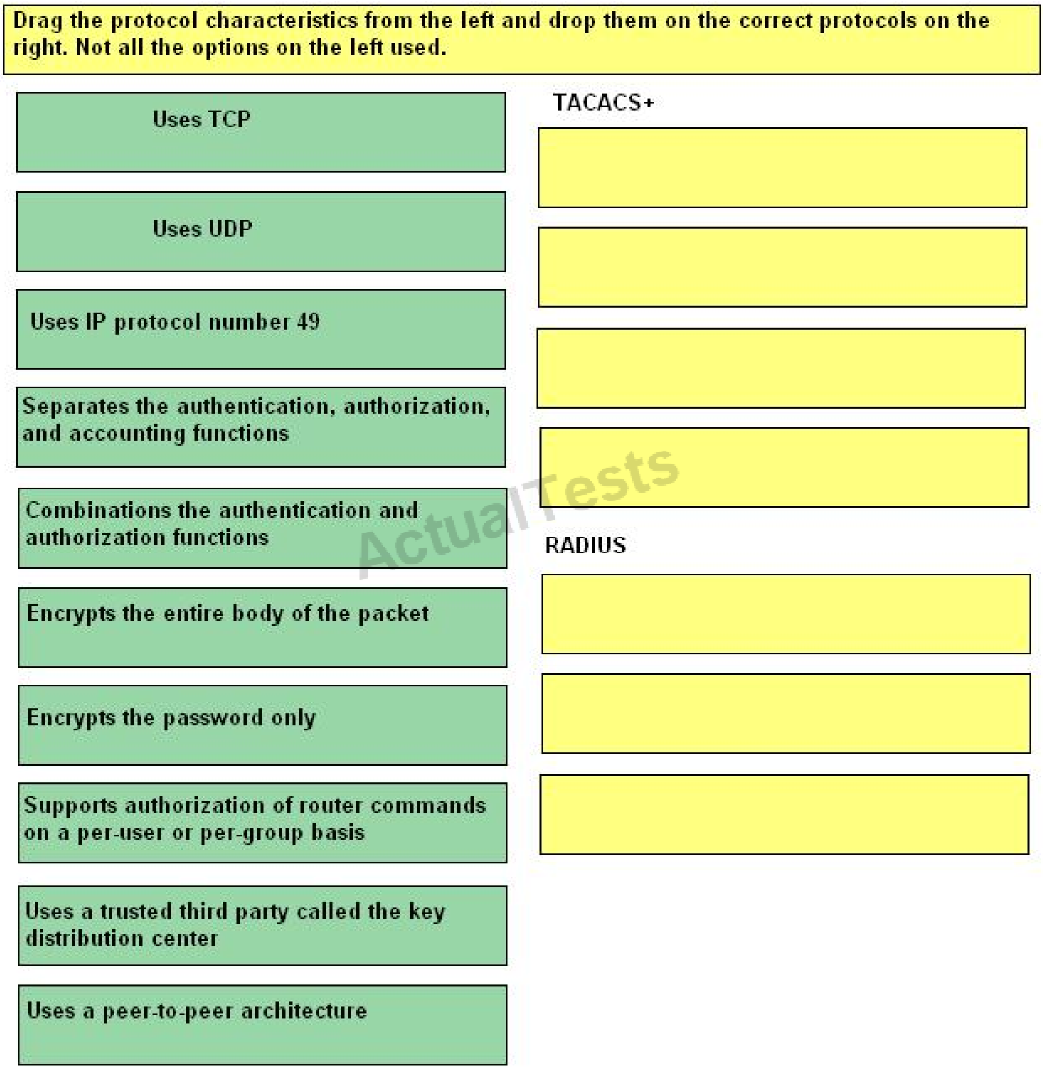
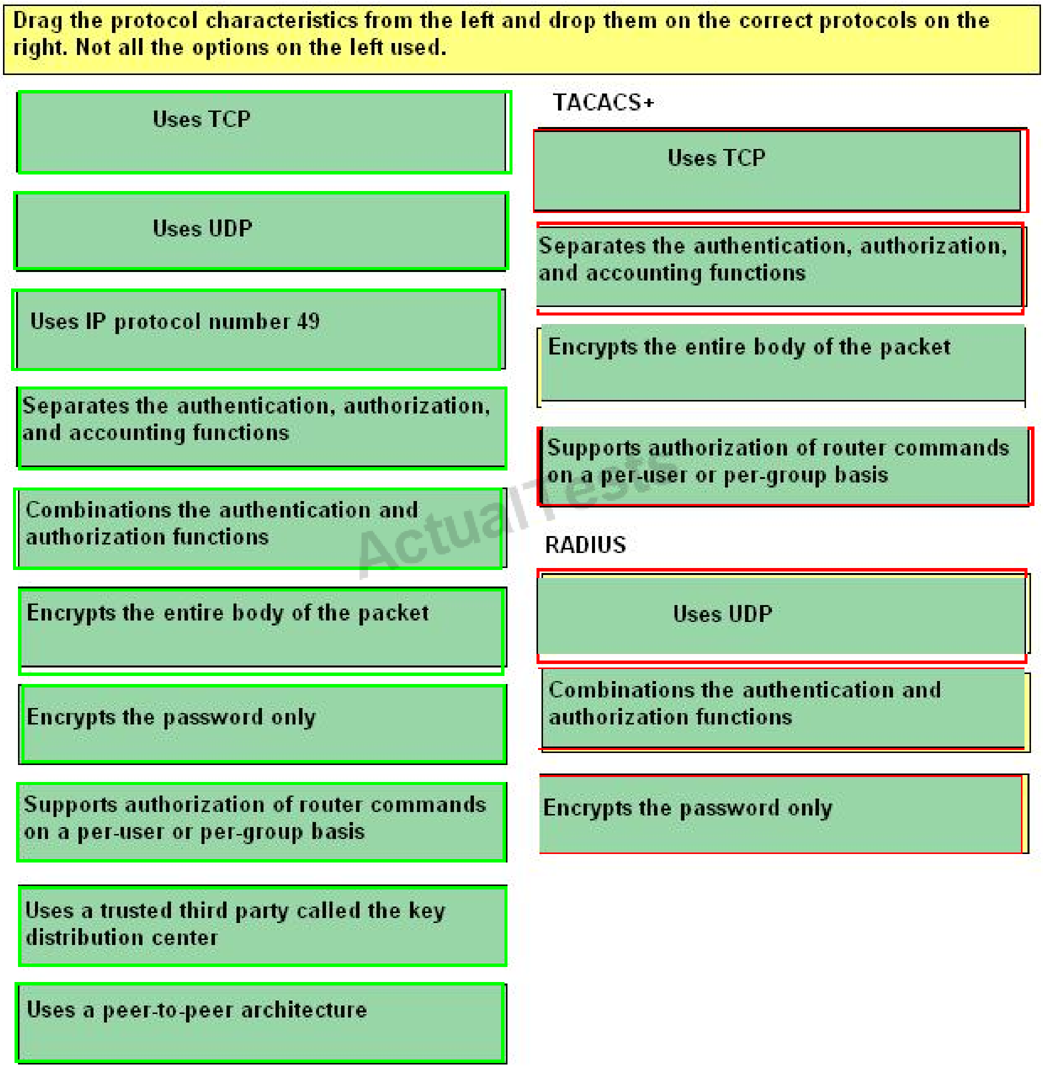
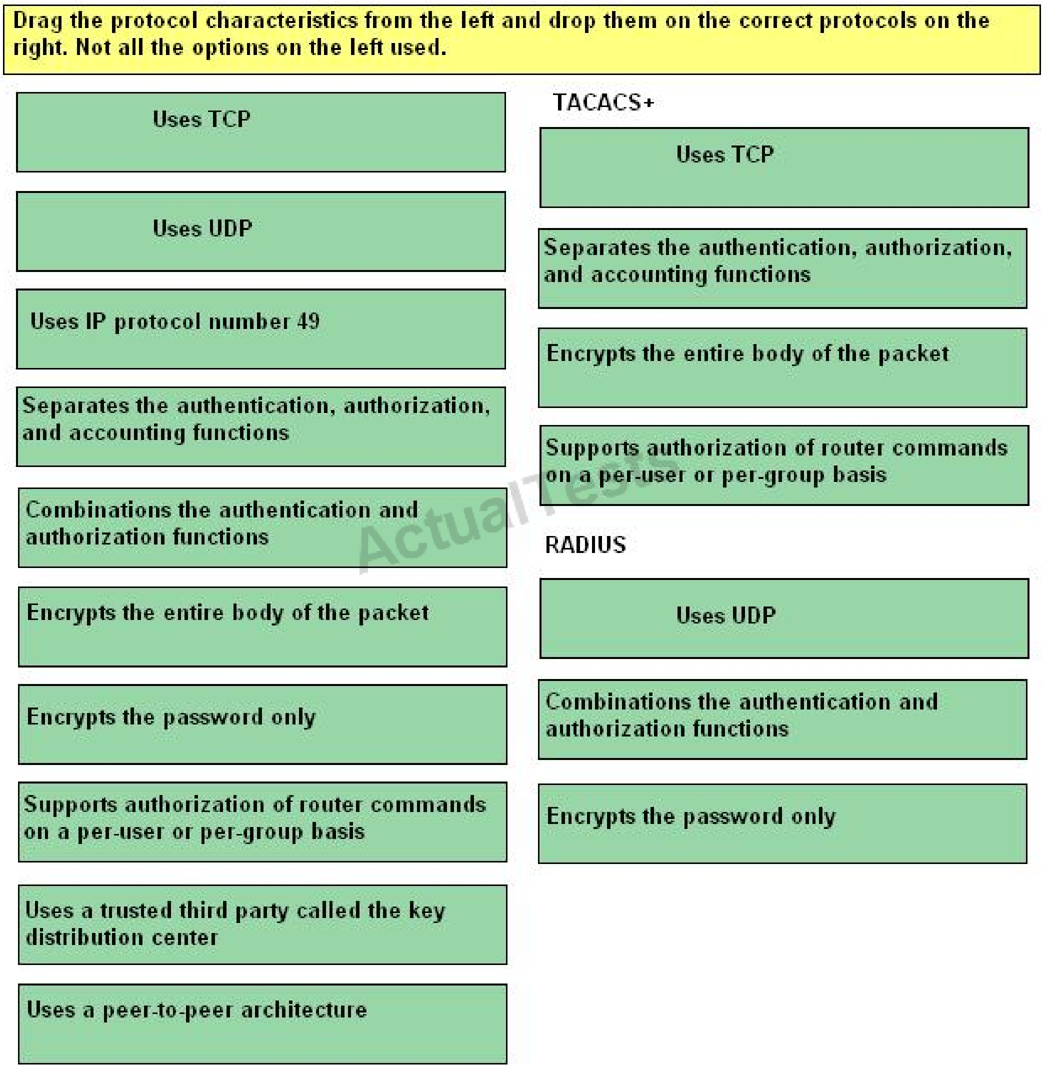
QUESTION NO: 105 DRAG DROP


Drag the protocol characteristics from the left and drop them on the correct protocols on the right. Not all the options on the left used.
Uses TCP
Uses UDP
Uses IP protocol number 49
Separates the authentication, authorization, and accounting functions
Combinations the authentication and authorization functions
Encrypts the entire body of the packet
Encrypts the password only
Supports authorization of router commands on a per-user or per-group basis
Uses a trusted third party called the key distribution center
Uses a peer-to-peer architecture
TACACS+
RADIUS

Answer:



Explanation:



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Reference: TACACS+ and RADIUS Comparison

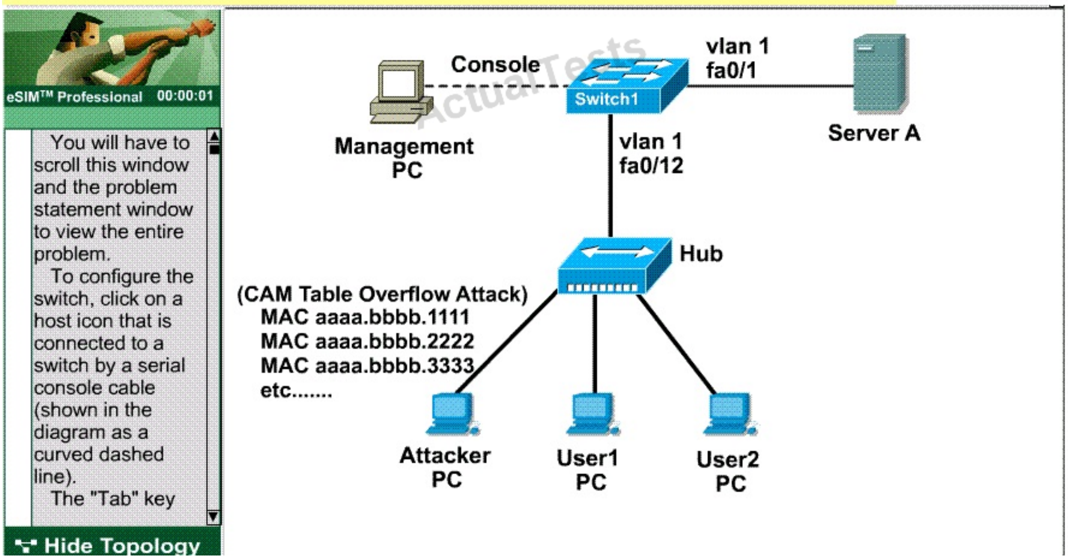
http://www.cisco.com/en/US/tech/tk59/technologies_tech_note09186a0080094e99.shtml#comp_packet_encry

QUESTION NO: 106 CORRECT TEXT

You are the network security administrator for Big Money Bank Co. You are informed that an attacker has performed a CAM table overflow attack by sending spoofed MAC addresses on one of the switch ports. The attacker has since been identified and escorted out of the campus. You now need to take action to configure the switch port to protect against this kind of attack in the future.

For purposes of this test, the attacker was connected via a hub to the Fa0/12 interface of the switch. The topology is provided for your use. The enable password of the switch is cisco. Your task is to configure the Fa0/12 interface on the switch to limit the maximum number of MAC addresses that are allowed to access the port to two and to shutdown the interface when there is a violation.

Enable password: cisco



```
-S#show run
Building configuration…
Current configuration :
!
version 12.1
no service pad
Service timestamps debug uptime
Service timestamps log uptime
No service password-encryption
!
Hostname    -S
!
!
Enable secret 5 $1$0/yw#toqA0XRiCtY8gh7pM06fS0
!
Ip subnet-zero
!
Ip ssh time-out 120
Ip ssh authentication-retries 3
!
Spanning-tree mode pvst
No spanning-tree optimize bpdu transmission
Spanning-tree extend system-id
!
……
!
Interface FastEthernet0/22
!
Interface FastEthernet0/23
!
Interface FastEthernet0/24
 switchport mode trunk
!
Interface GigabitEthernet0/1
!
Interface GigabitEthernet0/2
!
Interface vlan1
 ip address 172.26.26.202 255.255.255.0
 no ip route-cache
!
Ip http server
!
!
!
!
Line con 0
Line aux 0

Line vty 5 15
 password cisco
Login
!
End
   -S#
```


Answer: Switch1>enable

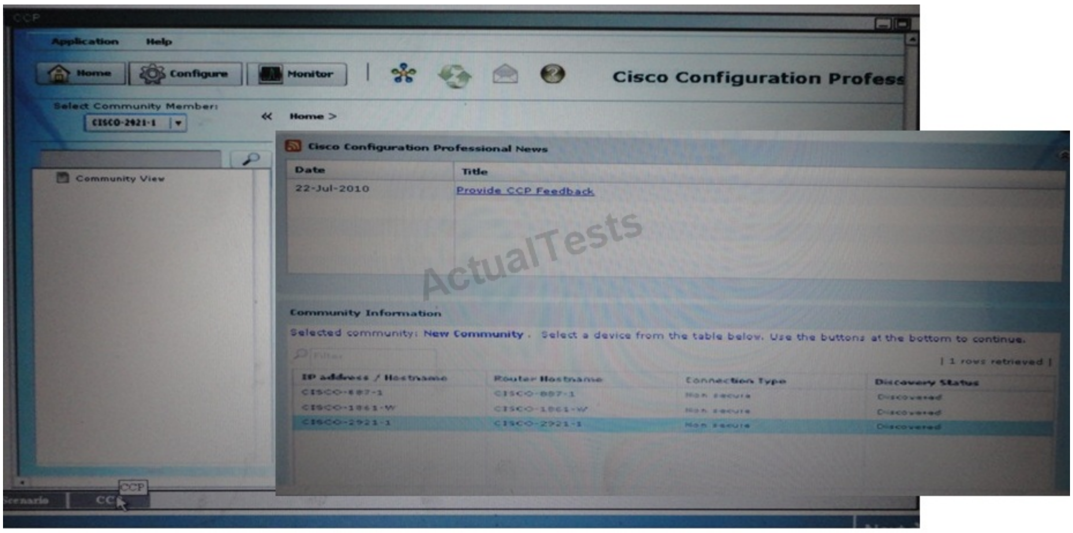
Switch1#config t

Switch1(config)#interface fa0/12
Switch1(config-if)#switchport mode access
Switch1(config-if)#switchport port-security maximum 2
Switch1(config-if)#switchport port-security violation shutdown
Switch1(config-if)#no shut
Switch1(config-if)#end
Switch1#copy run start

QUESTION NO: 107

Scenario:

You are the security admin for a small company. This morning your manager has supplied you with a list of Cisco ISR and CCP configuration questions. Using CCP, your job is to navigate the pre-configured CCP in order to find answers to your business question.



Which four properties are included in the inspection Cisco Map OUT_SERVICE? (Choose four)

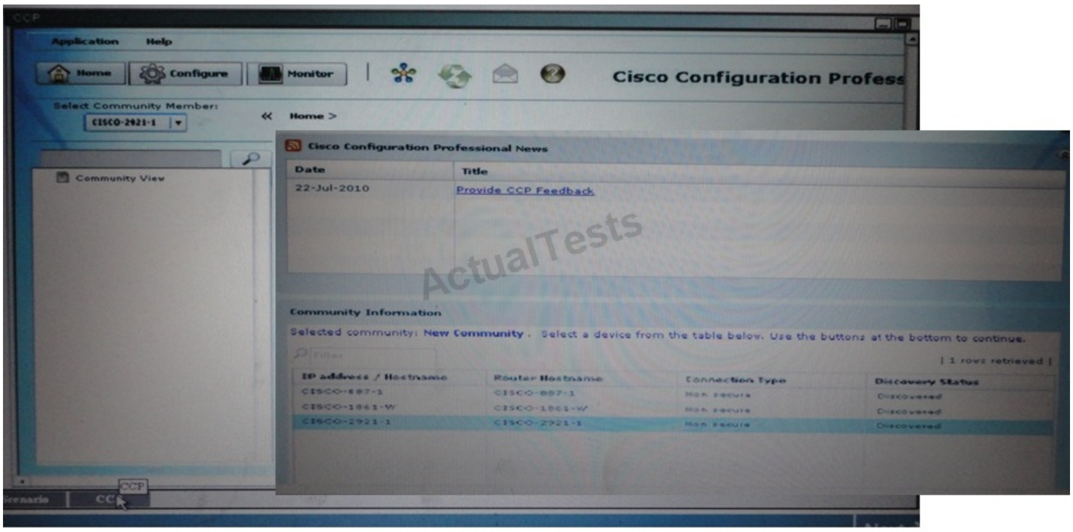
A. FTP
B. HTTP
C. HTTPS
D. SMTP
E. P2P
F. ICMP

Answer: A,B,E,F
Explanation:

QUESTION NO: 108

Scenario:

You are the security admin for a small company. This morning your manager has supplied you with a list of Cisco ISR and CCP configuration questions. Using CCP, your job is to navigate the pre-configured CCP in order to find answers to your business question.



What NAT address will be assigned by ACL 1?

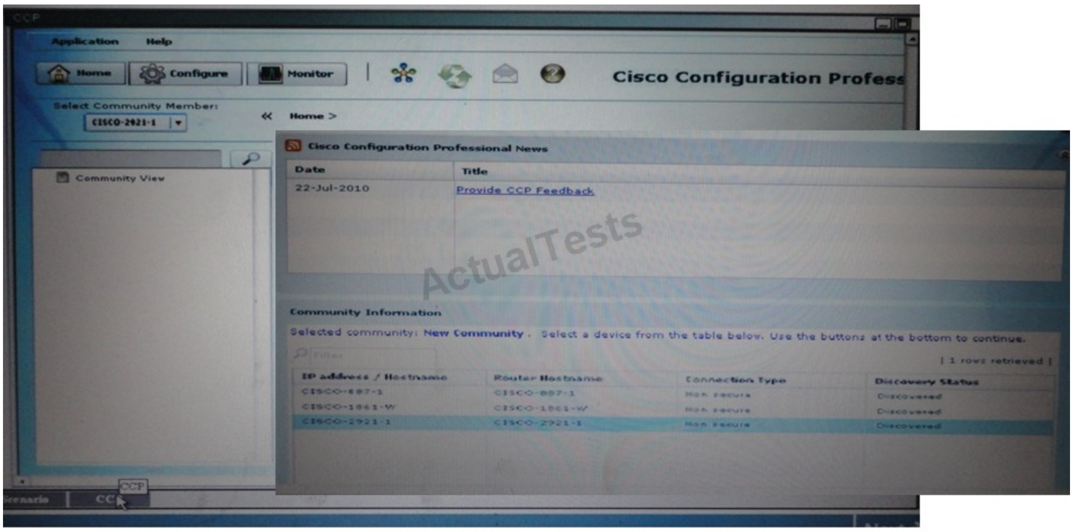
A. 192.168.1.0/25
B. GlobalEthernet0/0 interface address.
C. 172.25.223.0/24
D. 10.0.10.0/24

Answer: A
Explanation:

QUESTION NO: 109

Scenario:

You are the security admin for a small company. This morning your manager has supplied you with a list of Cisco ISR and CCP configuration questions. Using CCP, your job is to navigate the pre-configured CCP in order to find answers to your business question.



Which Class Map is used by the INBOUND Rule?

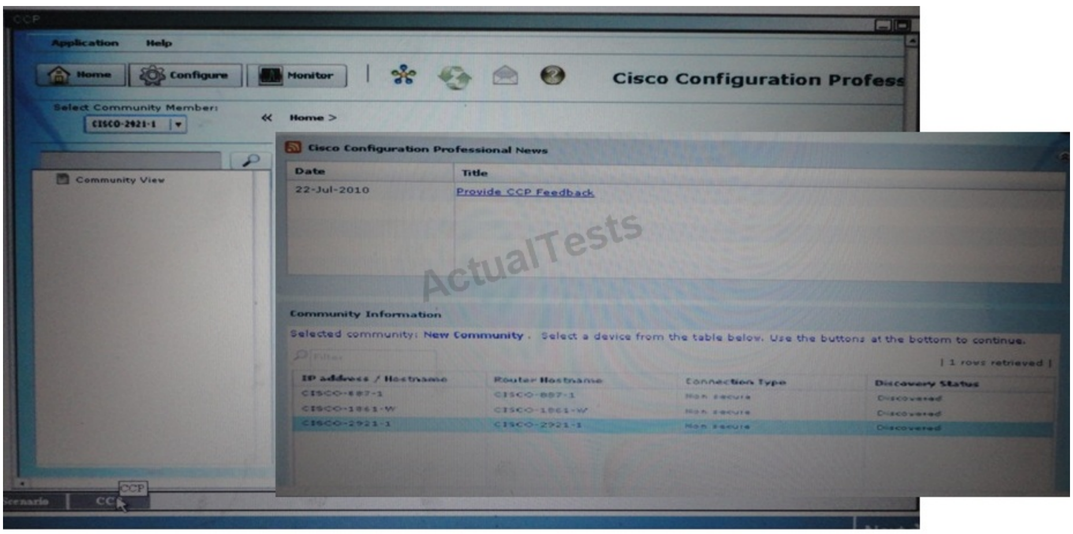
A. SERVICE_IN
B. Class-map-ccp-cls-2
C. Ccp-cts-2
D. Class-map SERVICE_IN

Answer: B
Explanation:

QUESTION NO: 110

Scenario:

You are the security admin for a small company. This morning your manager has supplied you with a list of Cisco ISR and CCP configuration questions. Using CCP, your job is to navigate the pre-configured CCP in order to find answers to your business question.



Which policy is assigned to Zone Pair sdm-zip-OUT-IN?

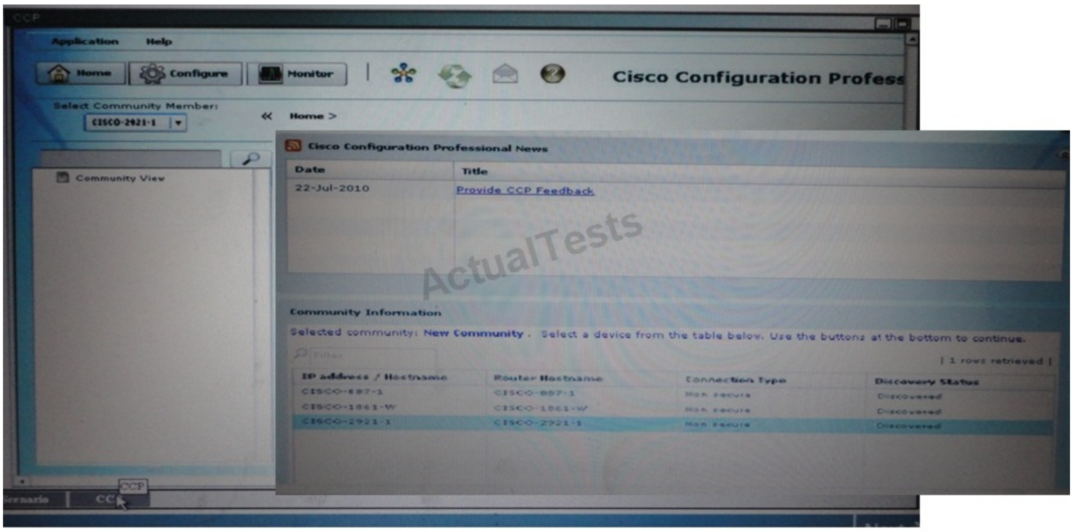
A. Sdm-cls-http
B. OUT_SERVICE
C. Ccp-policy-ccp-cls-1
D. Ccp-policy-ccp-cls-2

Answer: D
Explanation:

QUESTION NO: 111

Scenario:

You are the security admin for a small company. This morning your manager has supplied you with a list of Cisco ISR and CCP configuration questions. Using CCP, your job is to navigate the pre-configured CCP in order to find answers to your business question.



What is included in the Network Object Group INSIDE? (Choose two)

A. Network 192.168.1.0/24
B. Network 175.25.133.0/24
C. Network 10.0.10.0/24
D. Network 10.0.0.0/8
E. Network 192.168.1.0/8

Answer: B,C
Explanation: